Unraveling NFT Pricing Complexity

In-Depth Insights from Picture-for-Profile (PFP) Collections

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Introduction

Non-Fungible Tokens (NFTs) have emerged as a groundbreaking phenomenon with the potential to revolutionize various industries, particularly in the realm of digital assets. They are units of data stored on a blockchain certifying the uniqueness and distinctiveness of a digital asset and are reshaping the way we perceive digital creations in fields such as art, gaming, music, and collectibles. They have earned significant attention in early 2021. Within the first four months of 2021, the NFT market surpassed a trading volume of 2 billion USD.

On December 2, 2021, the market witnessed a historic sale with Pak's NFT artwork, "The Merge", fetching an astounding $91.8 million, further amplifying the significance and impact of NFTs in the art market and beyond. However, at the middle of 2022 the NFT market started facing some challenges, leading to a substantial decline in trading volume activity.

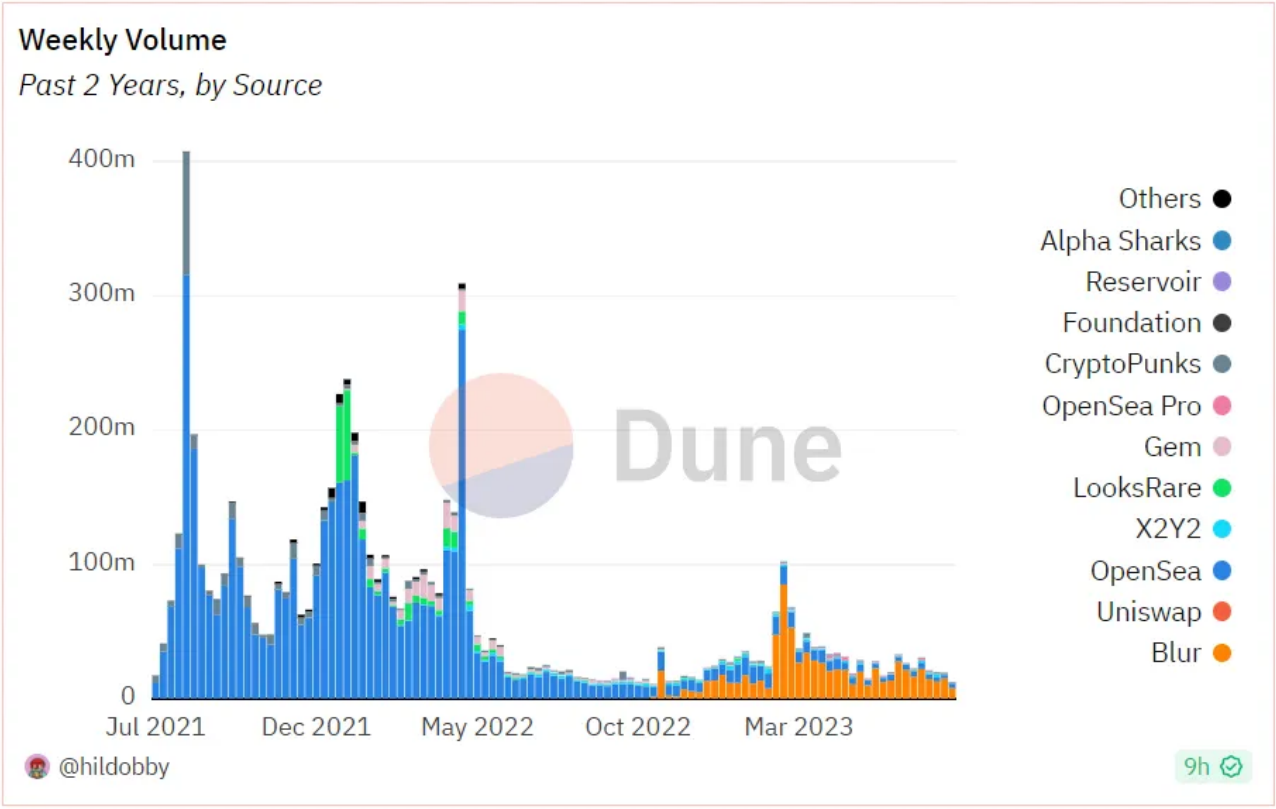


Figure 1: Weekly volume of the NFT market from different marketplaces

One category among all the spectrum of NFTs categories that has gained remarkable prominence is the picture-for-profile. PFPs represent digital avatars or profile images that users can adopt across various online platforms and social media networks. BAYC, one PFP collection, is held by a few notable NBA stars, singers and television hosts. Azuki's NFT profile pics sold out in three minutes and Okay Bears sold out all 10,000 NFTs in a day, bringing in a daily record of $18 million in sales volume.

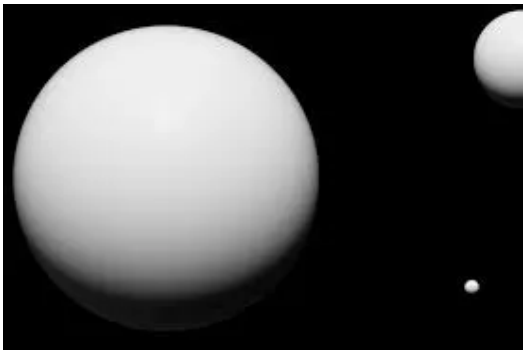


Figure 2: "The Merge" NFT

Focusing on this narrative, this research paper aims to contribute to the current knowledge landscape by conducting a thorough investigation into the intricate determinants influencing the pricing of NFTs, with a specific focus on

the picture-for-profile category. Through an in-depth analysis of five distinct NFT collections falling under this category, we seek to uncover the underlying internal and external factors that propel market dynamics and ultimately determine the intrinsic value of these digital assets.

In the subsequent sections of this research paper, we present the literature review, providing an overview of the current knowledge of NFTs, yet still limited, the methodology we used in this paper, as well as the database that we built, the results for our 5 analyzed PFP collections, the discussion of these results and the conclusion.

Literature Review

This literature review aims to explore the methodologies and findings of several studies that shed light on the multifaceted realm of NFTs. By delving into these different aspects, we aim to provide a comprehensive understanding of the NFT ecosystem, its complexities, and its potential implications.
NFTs, or non-fungible tokens, represent tradable rights that can be securely recorded as ownership tokens on the blockchain through the use of smart contracts (Ko et al. 2022). While there are various definitions of NFTs, a simplified explanation is that they are tokens that certify the uniqueness of digital assets (Kapoor et al. 2022).

However, in order to gain a better understanding of these digital assets, it is helpful to categorize them based on their different types. (Nadini et al. 2021) has identified six distinct groups of non-fungible tokens, namely Art, Collectible, Games, Metaverse, Utility and Other. Although this categorization provides a broad framework for discussing NFTs, the “Other” category is quite vague and lacks precision. To notice this limitation, we can cite (Mazur, n.d.), a research paper that delves into the risks and returns of NFTs and explores various financial use cases, such as trading, liquidity mining, farming, collateral-based loans, and fractional NFTs. These use cases, if classified, would all fall under the broader classification of “Other”, not being very precise.

Upon comprehending the categorization of NFTs and their use cases, we looked into the various factors that can influence the price of an NFT. In this regard, (Kaczynski and Kominers 2021) examined the creation of NFT value

through several key aspects; the scarcity, which stems from the unique nature of an NFT, making it highly desirable and valuable, the rarity, which is enhanced when some creators intentionally limit the number of tokens they produce, the authenticity, which involves the verification of an NFT's uniqueness, the provenance, which is related to the ownership history and transaction records of an NFT, the popularity, which plays a important role when the collection is associated with renowned artists or celebrities, or when it has gained significant attention on social media, and lastly, utility, which is an important factor to analyze whether an NFT offers more practical use cases beyond being a mere picture.

Furthermore, the influence of visual features in the price of NFT is a well-explored subject in the literature. For example, (Mekacher et al. 2022) thoroughly investigated the number of distinctive attributes linked to each NFT, as well as the distribution of these attributes within the collection. Building upon this idea, (Krasnoselskii, Madhwal, and Yanovich 2023) collected all attributes from an NFT collection, measuring their rarity and proposing a rarity score. In contrast, a different approach was taken by (Nadini et al. 2021) that considered the overall image of each NFT instead of analyzing individual attributes separately. Finally, (Alexander and Chen 2022) introduced the concept of grouping NFTs with similar attributes, creating clusters and conducting separate analyses for each cluster.

While these factors primarily encompass elements intrinsic to the NFT or its collection, there are also external factors that may influence the NFT's price. For example, (Borri, Liu, and Tsyvinski 2022) investigated the influence of the cryptocurrency market as well as traditional assets such as equity, commodities, and currencies on NFT prices.

Finally, we looked into various economic models to modelize the price of each NFT. When we have a number of observations on sales of objects that, although individually unique, have some degree of commonality, and trades are infrequent, all characteristics that we find in an NFT sale, we can apply the Hedonic Regression (Galbraith and Hodgson 2018). The Hedonic method was first proposed by (Court 1939) and then largely used and improved in the literature, with a focus on the Real Estate and Art market. (Agnello, n.d.) estimated the returns as well as the effects of various features of the paintings, like their size, who is the author, where it was auctioned, the medium of painting (e.g. oil, acrylic on canvas, panel, board, Masonite. . .) on price. (Meese and Wallace 1991) estimated house price for some cities in the United States of

America, considering for the feature variable the number of bathrooms and bedrooms, finished square footage, total number of rooms, dwelling age, and for the dummy variable the presence or absence of pools, fireplaces, assumability of mortgage, mortgage type and zoning type.

Besides, this model incorporates external factors and features that can affect house price, such as the impact of crime risk of property values (Linden and Rockoff 2008), the relationship between the undergrounding electricity and telecommunication network with house prices (McNair and Abelson 2010), and the impact of Noosa national park on surrounding property values (Pearson, Tisdell, and Lisle 2002).

However, it is important to note that it is not possible to include all potential characteristics that may influence the price of an asset. (Epple 1987) highlighted the issue of omitted variable bias, which poses a significant problem in hedonic regression models. To mitigate this problem, one approach is to adapt the hedonic model, which typically analyzes the price of a single asset, and instead consider a pair of sales. The repeat sales model, proposed and improved by (Case and Shiller 1987), is a powerful model that can estimate the house price index without explicitly considering the attributes of the house. An adapted version of the Case and Shiller index is currently utilized by the S&P 500 to calculate the house price index. While there is limited literature linking this method to the NFT market, (Borri, Liu, and Tsyvinski 2022) employs the repeat sales method to construct an overall index for the NFT market.

In parallel to the economic models, the emergence of machine learning and artificial intelligence has provided opportunities to develop ML models for predicting NFT prices. (Nadini et al. 2021) employed a deep convolutional neural network to extract the image feature and analyze their influence. Similarly, (Mekacher et al. 2022) used the same approach to identify latent variables that potentially drive the relationship between rarity and price. Moreover, (Jain, Bruckmann, and McDougall, n.d.) built a recurrent neural network to uncover the underlying factors affecting NFT prices, while (Alexander and Chen 2022) applied ML techniques, such as random forest algorithm, to predict the NFT rarity based on their attributes.

With that in mind, we can construct mathematical models to estimate the price of NFTs and find the factors that influence its valuation, but it is also important to consider alternative perspectives. According to (Kapoor et al. 2022), directly modeling NFT valuation as a mathematical economic system

may not be the most accurate approach. Instead, it should be viewed as a social phenomenon involving marketing schemes, the recognition and popularity of the NFT. The role of social media has a significant impact on that valuation and it emerges as a crucial factor of the NFT price. This perspective aligns with previous research that has examined the impact of Twitter on stock market prediction (Bollen, Mao, and Zeng 2011), highlighting the relevance of social media in financial contexts. Additionally, this same article references the Efficient Market Hypothesis, which suggests that stock markets are primarily influenced by new information, such as news, rather than relying solely on past and present prices. These insights encourage a holistic approach that incorporates social factors, market sentiment, and the timely incorporation of new information alongside mathematical modelling when analysing NFT valuations.

Empirical Methodology

Before delving into the mathematical model, we believed that it was essential to establish our own categorization framework to gain a comprehensive understanding of the NFT landscape, as it follows:
1) Social / Membership: NFTs that can grant access to exclusive events and communities.
2) Metaverse: NFTs that can represent plots of land on which it is possible to construct buildings or organize events. This digital space enables activities such as hosting online exhibitions or facilitating massive multiplayer video games.
3) Gaming: NFTs that are associated with any digital item from the domain of online games. It can be in-game items, characters, skins, maps, tickets, etc.
4) Proof of ownership of a specific physical or digital item: NFTs that have a physical artwork related in real life.
5) Financial NFTs: NFTs that represent fractional ownership of physical assets (real estate, cars. . .), stocks and bonds, futures and options contracts, virtual currency-backed NFTs.
6) Music: NFTs that represent albums or music releases.
7) Domain name: NFTs that can simplify decentralized domains, such as wallet addresses.
8) Pictures for profiles (PFPs): NFTs that were created to be displayed on social media profiles.
9) Collectibles: NFTs that have been created to be used and are attached to a project.
10) Photography: NFTs that represent unique photographs, often taken and edited digitally.

11) Comics: NFTs that represent unique comic book pages, often featuring digital art and storytelling
12) 3D Sculptures: NFTs that represent unique 3D sculptures, created using 3D modeling software and often animated or interactive.
13) Digital Paintings: NFTs that represent unique digital paintings, created using digital tools and softwares.
14) Artistic Videos: NFTs that represent unique video works of art, often including animation, motion graphics, and sound.
15) Mixed Media Art: NFTs that represent unique digital pieces that combine multiple types of media, such as painting, sculpture, photography, and digital design.

With the aid of this categorization framework, we can develop a comprehensive overview of the NFT ecosystem, facilitating analysis of the potential factors that may impact its pricing. We have classified these factors into two distinct categories: internal and external.

Internal Factors:

1) Artist Popularity/Brand Recognition: This variable refers to the fame and recognition of the artist or creator behind the NFT. The more popular or well-known the creator, the more valuable the NFT may be.
2) Quality of Artwork/Visual Features: This variable refers to the aesthetic value of the artwork or visual features of the NFT. The higher the quality and appeal of the artwork or visual features, the more valuable the NFT may be. This can also be seen as the value or appeal of the artwork from a market perspective.
3) Number of NFTs (Scarcity): This variable refers to the number of NFTs available in circulation. The scarcer the NFT collection is, the more valuable it may be.
4) Rarity: This variable is similar to the previous one and refers to how unique the characteristics in the NFT are. The more unique the attributes in an NFT, the more valuable it may be.
5) Blockchain Technology: This variable refers to the blockchain chosen for creating and maintaining the NFT. The more popular and widely used the chosen blockchain, the more valuable the resulting NFT may become.
6) Added Utility: This variable refers to any additional benefits or uses that

the NFT may have beyond its intrinsic value. For example, if the NFT grants access to exclusive content or events, it may be more valuable.
7) Historical Significance (OG NFTs): This variable refers to the historical significance of the NFT, especially if it was one of the first NFTs ever created. The more historical significance the NFT has, the more valuable it may be.
8) Process: This variable refers to the process of creating a collection with respect to the various methods used by the artist.

External Factors:

1) Social Media: This variable refers to the impact that social media may have on the value of the NFT. If the NFT is popular on social media platforms, it may be more valuable.
2) Market Trends/Demand: This variable refers to the current market trends and demand for NFTs. If there is a high demand for NFTs, the value of the NFT may increase.
3) Economic Conditions: This variable refers to the overall economic conditions, such as inflation, that may affect the value of the NFT. For example, during an economic downturn, the value of NFTs may decrease.

By incorporating both internal and external factors, we aim to gain an extensive understanding of the various elements that can shape the value and desirability of NFTs within the studied collections. We gathered the data focusing on the most prominent PFP collections in terms of volume, namely CryptoPunk , Bored Ape Yacht Club, Mutant Ape Yacht Club, Azuki, CloneX, and Moonbirds. We opted to exclude CryptoPunk from our analysis because it was one of the first NFT collections launched in 2017, before the implementation of the ERC-721 token protocol that is used nowadays. Consequently, relevant historical sales data for that collection was not found, leading to its omission from our research.

Initially, we turned to Etherscan, an Ethereum block explorer and analytics platform, to obtain the data, since all the collections mentioned are on the Ethereum blockchain. While etherscan provided information on sales and minting, and we got the mint data for our analysis, we encountered an issue when sales were conducted in currencies other than ETH, not being accurately represented. Furthermore, the traits data for each NFT was not available on Etherscan, leading us to search for another option.

Our next approach involved exploring Alchemy, a web3 developer platform, to find the desired data. Unfortunately, we were unable to retrieve the traits data for the collections, prompting us to quickly dismiss this option. Consequently, we continued our search for a reliable source of NFT data. Although Reservoir had the historical sales data, we could not rely on that third option due to the presence of missing data when manually verifying the information.

Ultimately, we turned to NFTPort, where we successfully obtained all the desired data except for the mint price and mint date, which we got on Etherscan. For each collection, we retrieved the traits of all the NFTs, the historical sales data for each token ID, as well as the buyer and seller addresses. Additionally, we obtained the sale prices in both ETH and USD, although, in some instances, only the ETH price was available. To address this, we utilized the historical price of that token to calculate the USD price, available at www.coingecko.com/en/coins/ethereum.

We made a slight adjustment to the historical sales data to suit our model's requirements. Specifically, we observed that some NFTs were sold multiple times on the same day with price differences of less than 5%. Such cases indicated that these sales were driven purely by speculative motives. To streamline our dataset and eliminate potential noise from speculative transactions, we decided to retain only the first sale of each NFT on a given day, excluding any subsequent sales. In addition to the NFT data, we also gathered various time series for further analysis, such as the S&P 500 index from Yahoo Finance, the Federal Funds Rate available at https://www.newyorkfed.org/markets/reference-rates/effr, and the ETH price data from CoinGecko.

As mentioned in the literature review, we modeled our problem as a hedonic model (Court 1939):

$$\ln p_{i,t} = \alpha_0 + \sum_{N=features} \beta_N B_{i,N} + \sum_t \gamma_t D_{i,t} + \epsilon_{i,t}$$

where:

$p_{i,t}$ is the price of sale of the ith NFT sold at the time t

α_0 is the intercept term

$B_{i,N}$ is the vector of features of the ith NFT. If the ith NFT has the trait N, the vector has a value equal to 1 and 0 otherwise

β_N is the regression coefficient that represents impact of a specific trait in the sale price

$D_{i,t}$ is the time vector that records the year of sale (it takes 1 if the NFT was sold in the time t)

γ_t is the regression coefficient that represents the change in NFT prices between two periods.

$\epsilon_{i,t}$ is the residual term that accounts for the variability in the dependent variable that is not explained by the independent variables

Before proceeding with the model, we first addressed the problem of multicollinearity, which occurs when two or more independent variables are highly correlated, making it challenging to isolate their individual effects on the dependent variable. To tackle this problem, we calculated the correlation matrix for all independent variables in the dataset. This matrix reveals the pairwise correlation between each pair of independent variables, ranging from -1 to +1, where -1 indicates a perfect negative correlation, +1 represents a perfect positive correlation, and 0 suggests no correlation.

Next, we established a threshold of ±0.7 for the correlation coefficient, and if the correlation between two variables exceeded this threshold, we excluded one of them from the model. By doing so, we retained the most relevant and informative predictors while mitigating the impact of multicollinearity.

To determine the coefficients of the Hedonic model, we adopted the Ordinary Least Squares method (OLS), which minimizes the sum of the squared differences (residuals) between the observed dependent variable and the predicted values generated by the linear equation.

It is worth noting that our NFT's characteristics remain constant over time, and thus our vector of features, denoted Bi,N , is the same for each period. Greenlees (1982), Mark and Goldberg (1984) analyzed this special case when the feature vector varies over the time, but we don't need to treat our problem in this way.

Following the coefficient estimation, we examined the correlation and causality between the regression coefficient related with the time, denoted as γt, and three other time series: S&P 500 index, FED rate and ETH price.

To measure the correlation between two series, we utilized the Pearson Test, which is a statistical test to assess the strength and direction of two variables, by calculating the Pearson correlation coefficient (r). This coefficient ranges from -1 to +1, where -1 represents a perfect negative linear relationship, +1 represents a perfect positive linear relationship, and 0 indicates no linear relationship.

For measuring causality between two series, we employed the Granger Test, which is a statistical test that helps to determine whether one time series can predict another time series, revealing the temporal relationship between variables and assessing the predictive power of past values from one variable on another one.

Results

Bored Ape Yacht Club

Proceeding with the methodology described in the previous section, we started by analyzing the correlation between all the traits. For the Bored Ape Yacht Club, we didn't find any correlated traits (with the correlation matrix coefficient higher than 0.7).

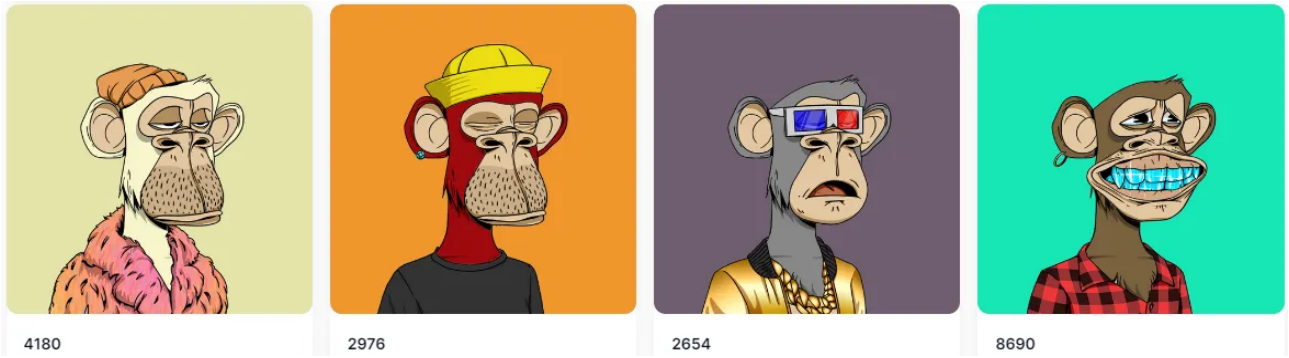


Figure 3: Example of BAYC NFTs

Following the steps, we ran our Hedonic Model to find the appropriate model to predict the collection price, considering only the statistically significant coeffi-

cients. We found that all coefficients were statistically significant at a significant level of 0.05, but the coefficients related to the background feature.

Model BAYC
$\ln \hat{p_{i,t}} = \hat{\alpha_0} + \sum_{N=traits} \hat{\beta_N} B_{i,N} + \sum_t \hat{\gamma_t} D_{i,t}$

Table 1: Hedonic model with significant coefficients for BAYC

We continued the analysis by measuring the linear relationship between the estimated coefficients and its rarity and finally we ran a Pearson test and a Granger test to verify the correlation and causality of the temporal estimated coefficient and three other temporal series.

Pearson correlation test	Granger causality test
ETH, S&P500, Fed Rate	No causality conclusion

Table 2: Pearson and Granger test for BAYC

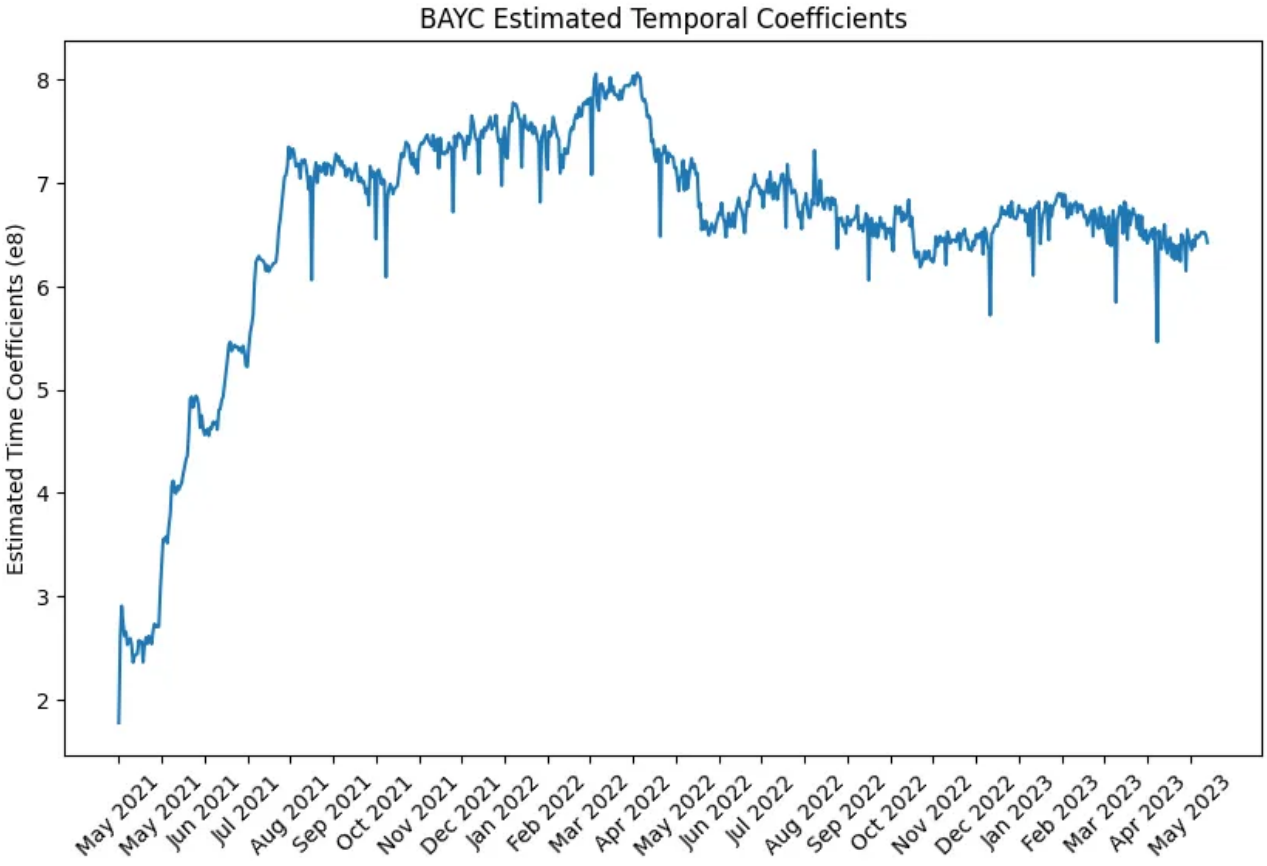


Figure 4: BAYC estimated temporal coefficients from May 2021 to May 2023

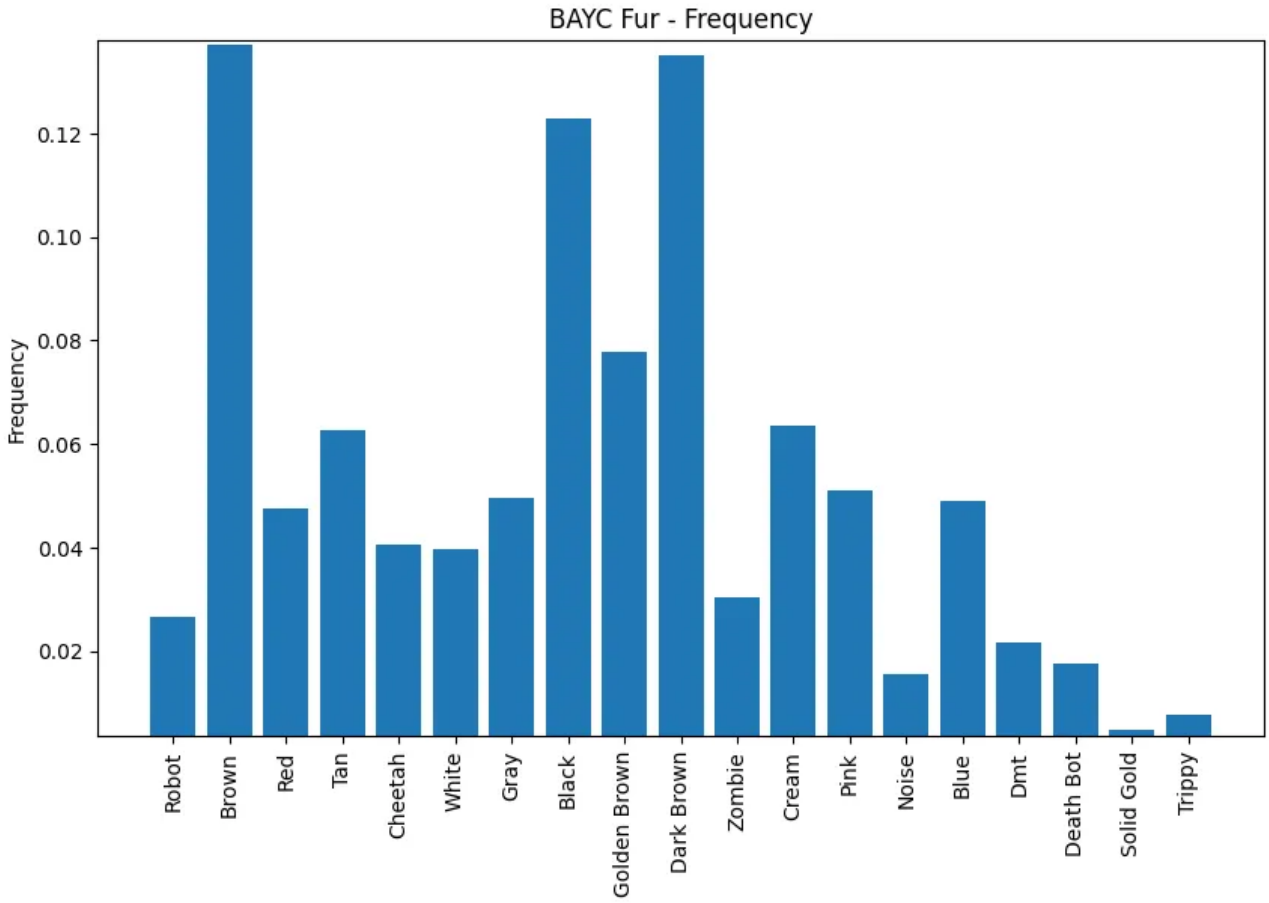


Figure 5: Frequency of fur trait within the BAYC collection

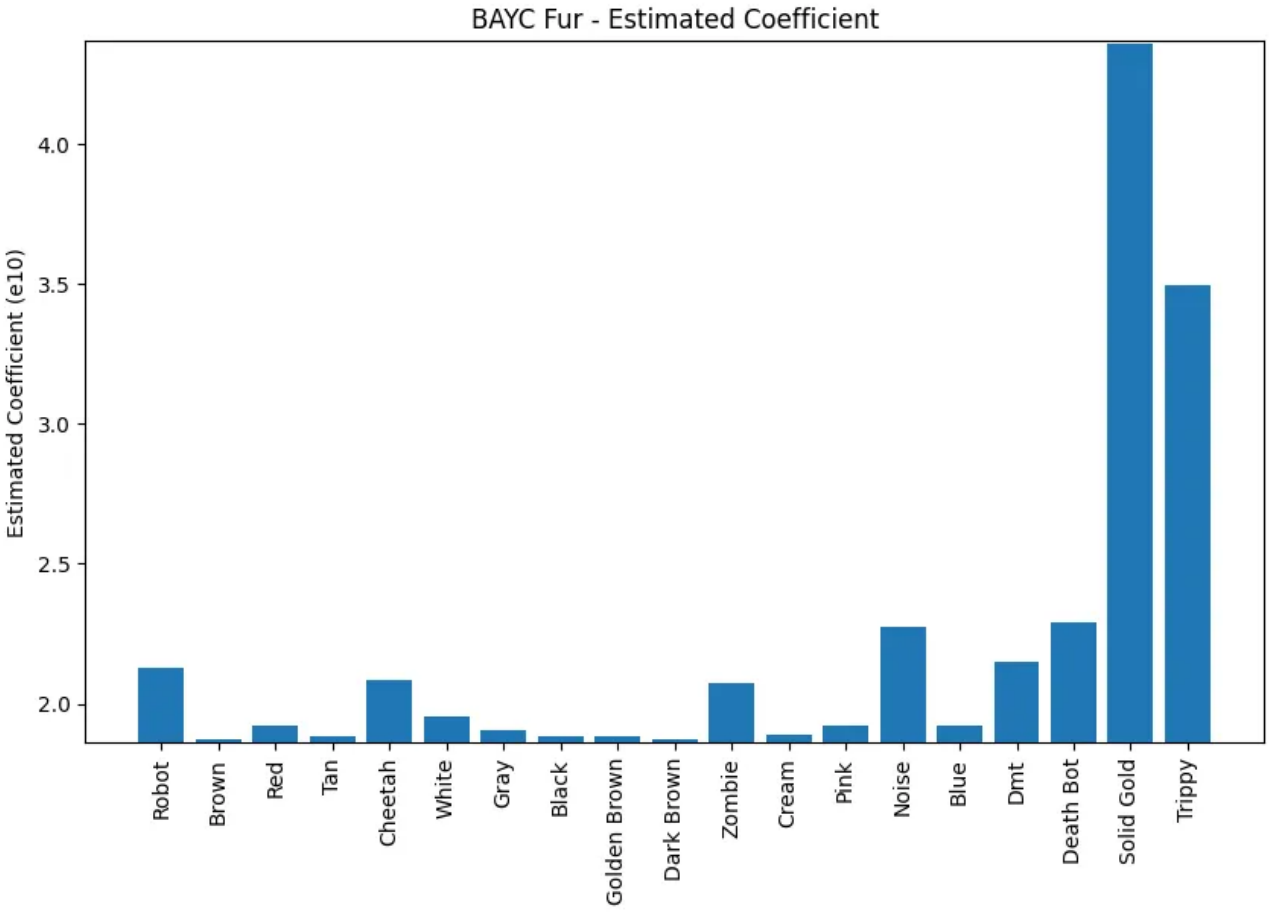


Figure 6: Fur estimated coefficients for BAYC

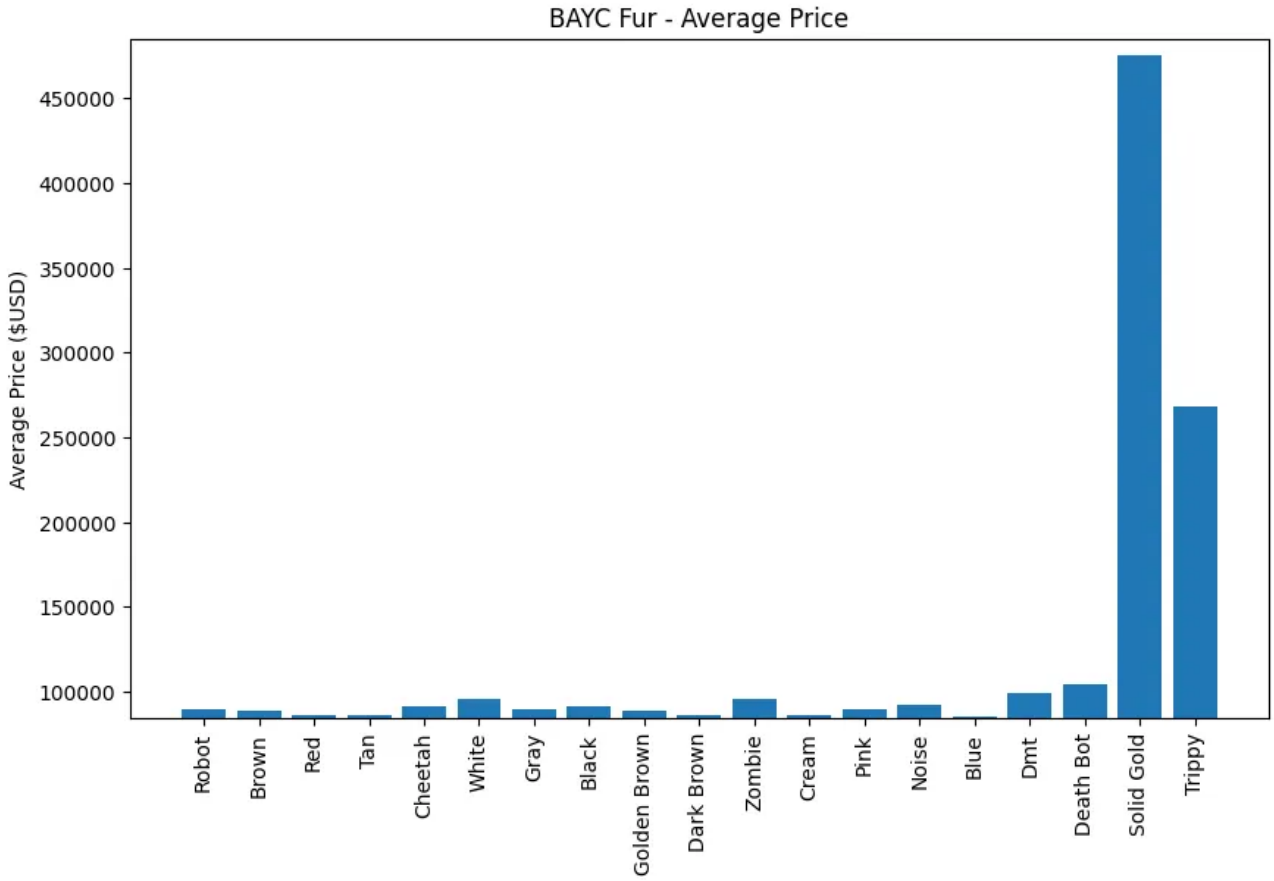


Figure 7: Average sale price of different types of fur within the BAYC collection

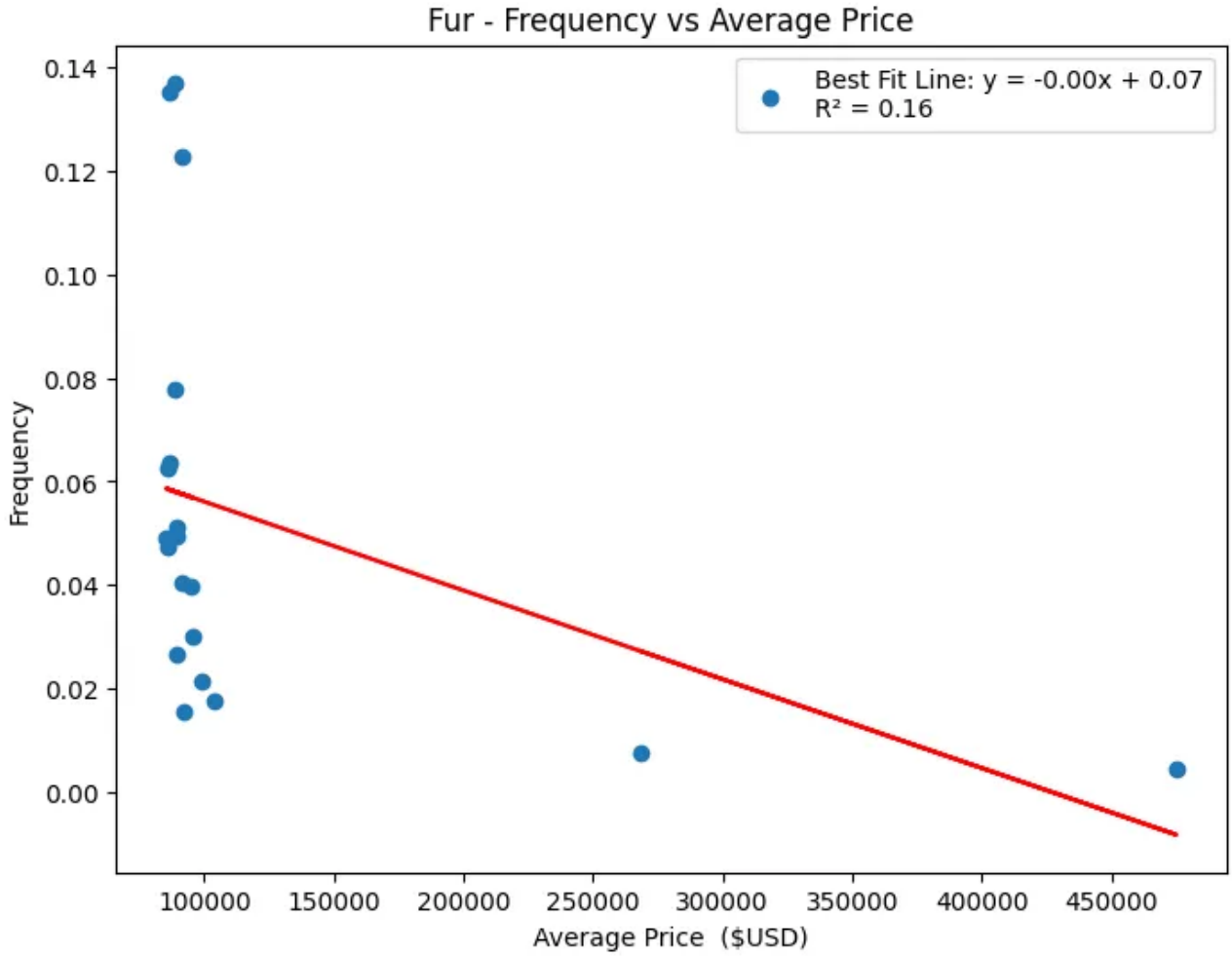


Figure 8: Linear regression of the fur frequency on the fur average price

Mutant Ape Yacht Club

This collection is a particular one because it has a different type of mint for some specific token ids: the last 20000 NFTs can only be minted if you have a BAYC NFT (a detailed explanation will be given in the next section). Our first approach was to treat all 30000 NFTs as equal.

We didn't find any statistically significant coefficient.

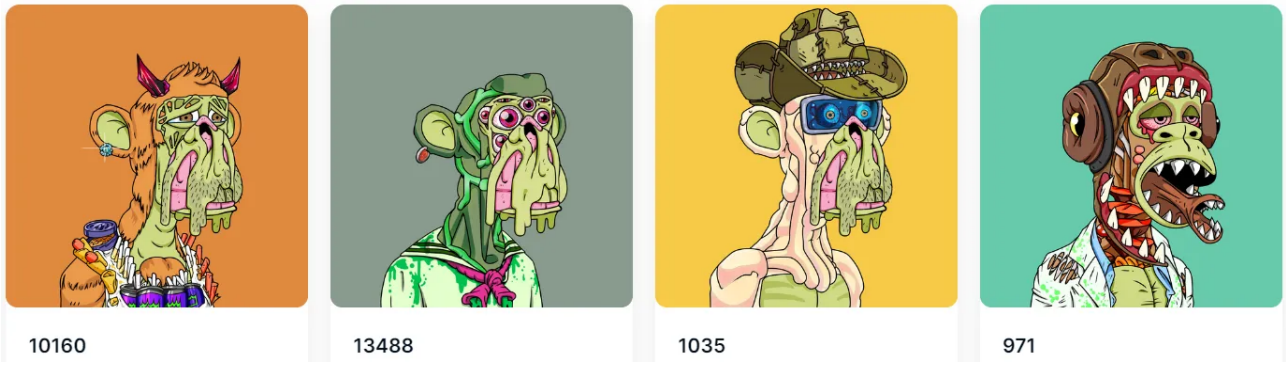


Figure 9: Example of MAYC NFTs

As a second approach, we divided the collection into two: the first being the 10000 NFTs (Token ID 0 to 9999) and the next 20000 (token id 10000 to 29999). The explanation of that division will be given in the next section.

For the first group, we found only 8 significant coefficients and we didn't continue the analysis for this first batch.

For the second group, we found the following model, with 764 statistically significant coefficients, but with 260 not statistically significant coefficients.

Model MAYC (token id from 10000 to 29999)
$\ln \hat{p_{i,t}} = \hat{\alpha_0} + \sum_{N=traits} \hat{\beta_N} B_{i,N} + \sum_t \hat{\gamma_t} D_{i,t}$

Table 3: Hedonic model with significant coefficients for MAYC 10000-29999

Because of the high number of statistically insignificant coefficients, we decided to not proceed the analysis for the second sample either.

Azuki

For the Azuki collection, we found 14 pairs with the correlation matrix coefficient higher than 0.7 and thus we excluded one of each pair from our feature vector.

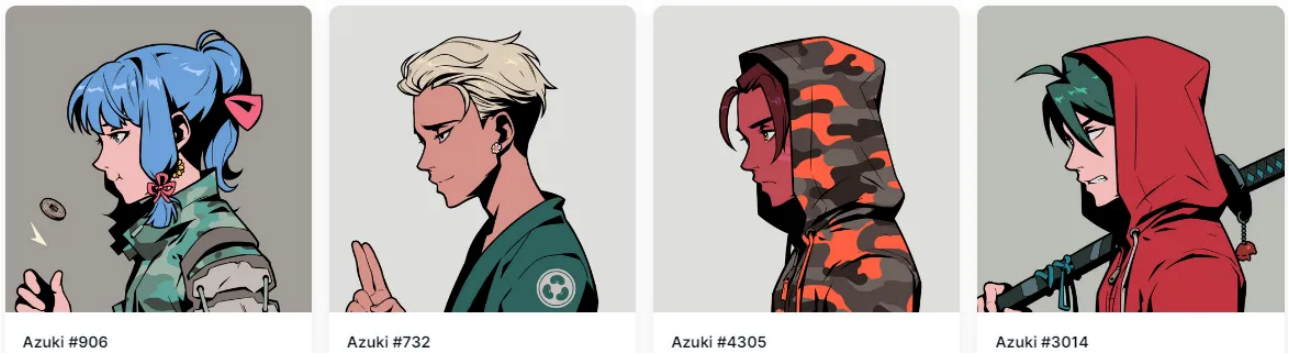


Figure 10: Example of Azuki NFTs

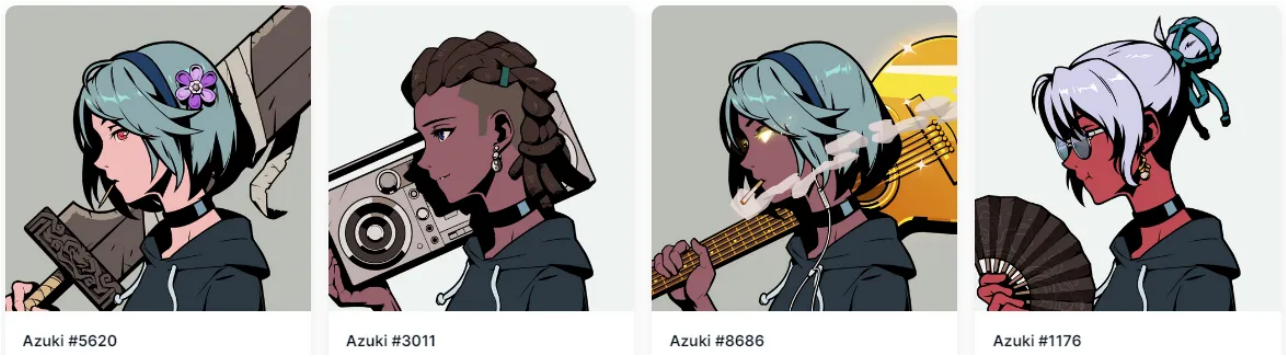


Figure 11: Example of correlation: all clothing type ”Hoodie” have a neck type ”Choker”

For the Hedonic Model, at a significant level of 0.05, we found that only the temporal coefficients are significant to predict the price.

Model Azuki
$\ln \hat{p_{i,t}} = \sum_t \hat{\gamma_t} D_{i,t}$

Table 4: Hedonic model with significant coefficients for Azuki

Following with the Pearson test and Granger test, we found that the estimated temporal coefficients are correlated with the ETH price, S&P500 and with the Fed Rate, and we found a causality with the S&P500 index with lag 2, 3, 4, and 5.

Since we didn’t have significant coefficients for the features, we didn’t verify the relationship between them and rarity.

Pearson correlation test	Granger causality test
ETH, S&P500, Fed Rate	S&P500 (lag 2, 3, 4, and 5)

Table 5: Pearson and Granger test for Azuk

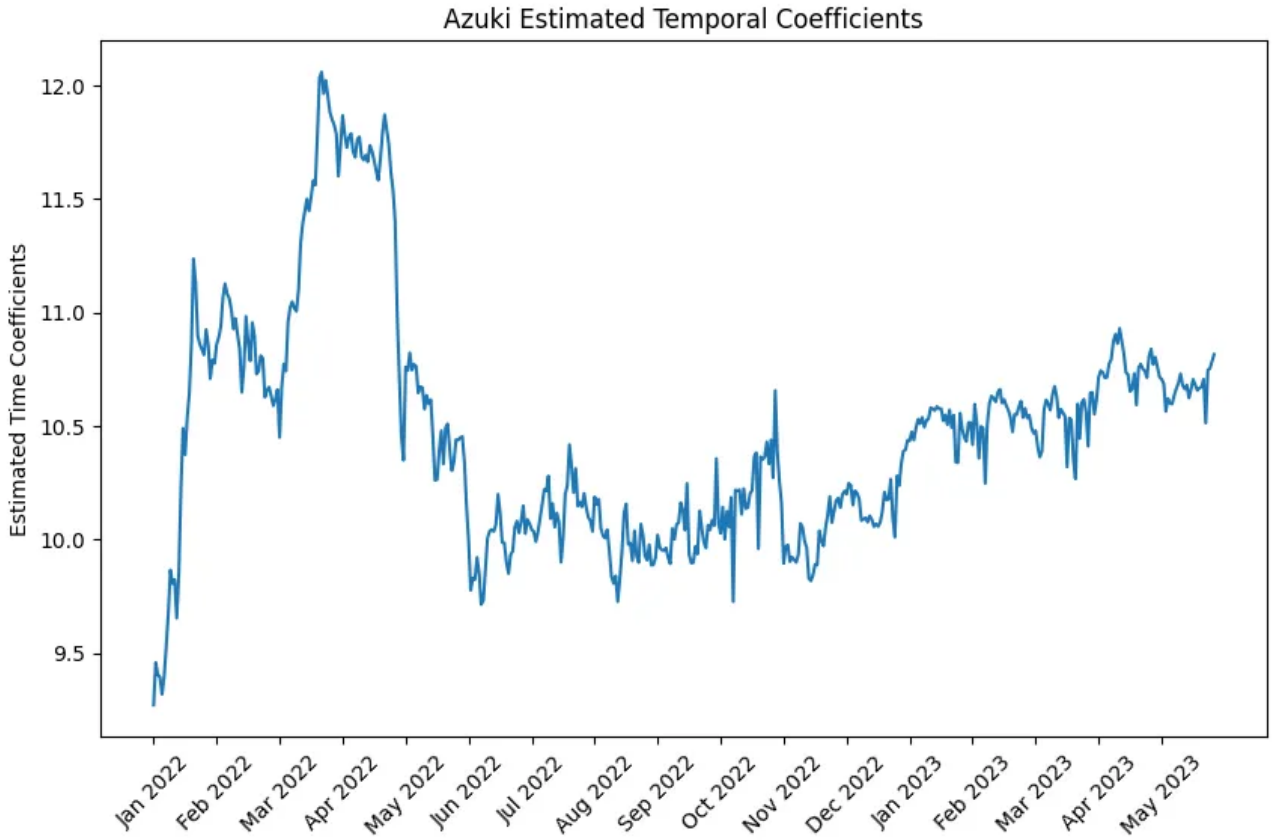


Figure 12: Azuki estimated temporal coefficients from Jan 2022 to May 2023

CloneX

For the CloneX collection, we found 8 correlated pairs of features and we excluded one feature of each pair from our feature vector.

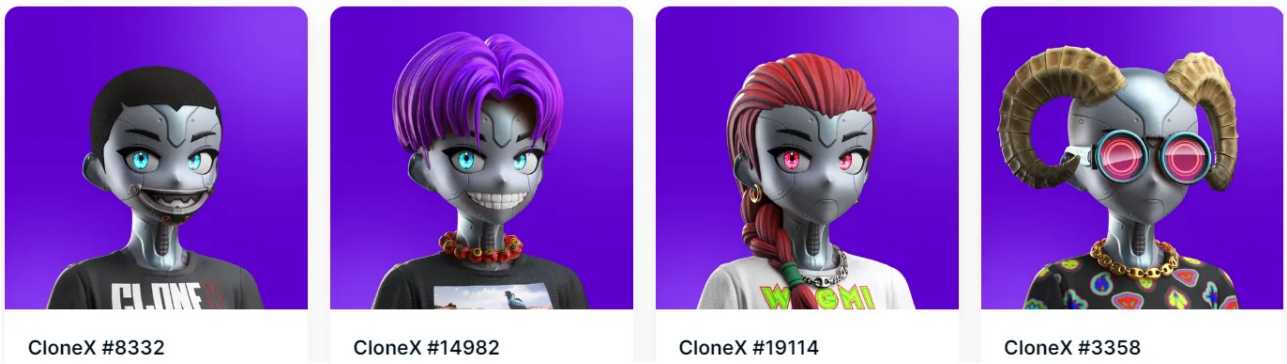
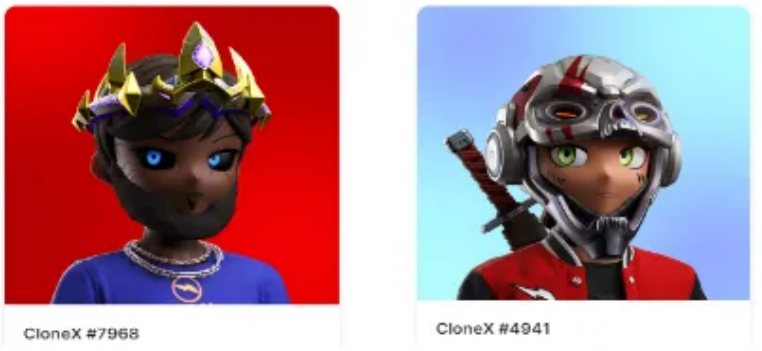


Figure 13: Example of CloneX NFTs



(a) The hair "CHOSEN 1 CRWN" is correlated with the mounth "LBJ Beard"

(b) The clothing " Blck-rd varsity jckt" is correlated with the helmet "Skull"

Figure 14: Two examples of correlated traits in the CloneX collection

For this collection, we didn't have a sufficient number of statistically significant coefficients. Only for these cases of features (Table 6).

Since we didn't have enough coefficients, we didn't proceed with the Granger test, nor the Pearson test.

CloneX - Significant coefficients at 0.10
Eye Color: YLLW Breezy, RNBW, AWAKEN, SPINNING, RD CRYSTL, HALF-OPEN, BLU CRYSTL, VVS CRYSTL, CLSD YLLW Breezy, CLSD RNBW, CLSD BLU CRYSTL, CLSD RD CRYSTL, CLSD VVS CRYSTL
Jewelry: WGLD VVS CLSP

Table 6: Significant coefficients for the CloneX collection

Moonbirds

For the Moonbirds collection, we found 16 correlated pairs of features and we excluded one feature of each pair from our feature vector.



Figure 15: Example of Moonbirds NFTs

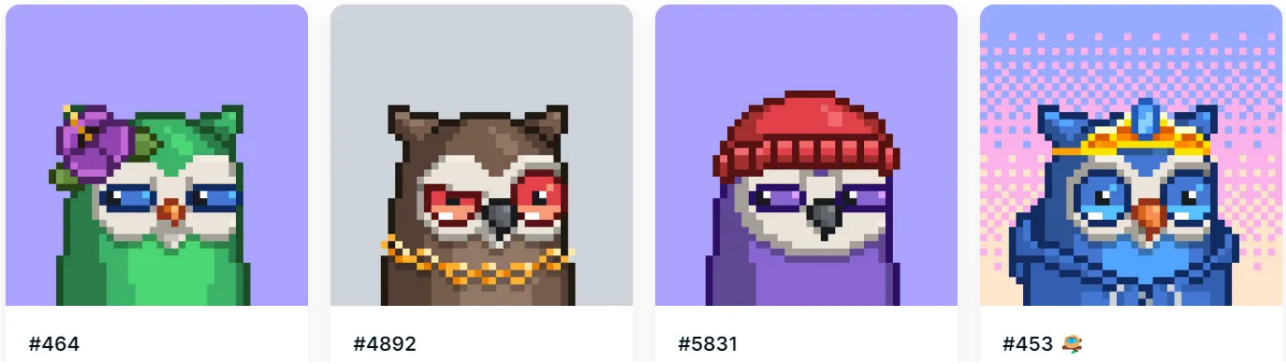


Figure 16: Example of correlation: all Body type ”Robot” have feathers type ”Metal”

For the model, at a significance level of 0.05 we didn't have any statistically significant coefficients, but at 0.10 we found the following model:

Model Moonbirds
$\ln \hat{p_{i,t}} = \hat{\alpha_0} + \sum_{N=traits} \hat{\beta_N} B_{i,N} + \sum_t \hat{\gamma_t} D_{i,t}$

Table 7: Hedonic model with significant coefficients for Moonbirds

Some traits were not statistically significant (Table 8).

We continued the analysis by measuring the linear relationship between the estimated coefficients and its rarity and finally we ran a Pearson test and a Granger test to verify the correlation and causality of the temporal estimated coefficient and three other temporal series.

Moonbirds - Not Significant coefficients at 0.10
Background: Green, None, Purple, Yellow, Blue, Pink, and Gray
Beak: Small, Long, and Short
Feathers: None

Table 8: Not significant coefficients for the Moonbirds collection

Pearson correlation test	Granger causality test
ETH, S&P500, Fed Rate	ETH Price (lag 5)

Table 9: Pearson and Granger test for Moonbirds

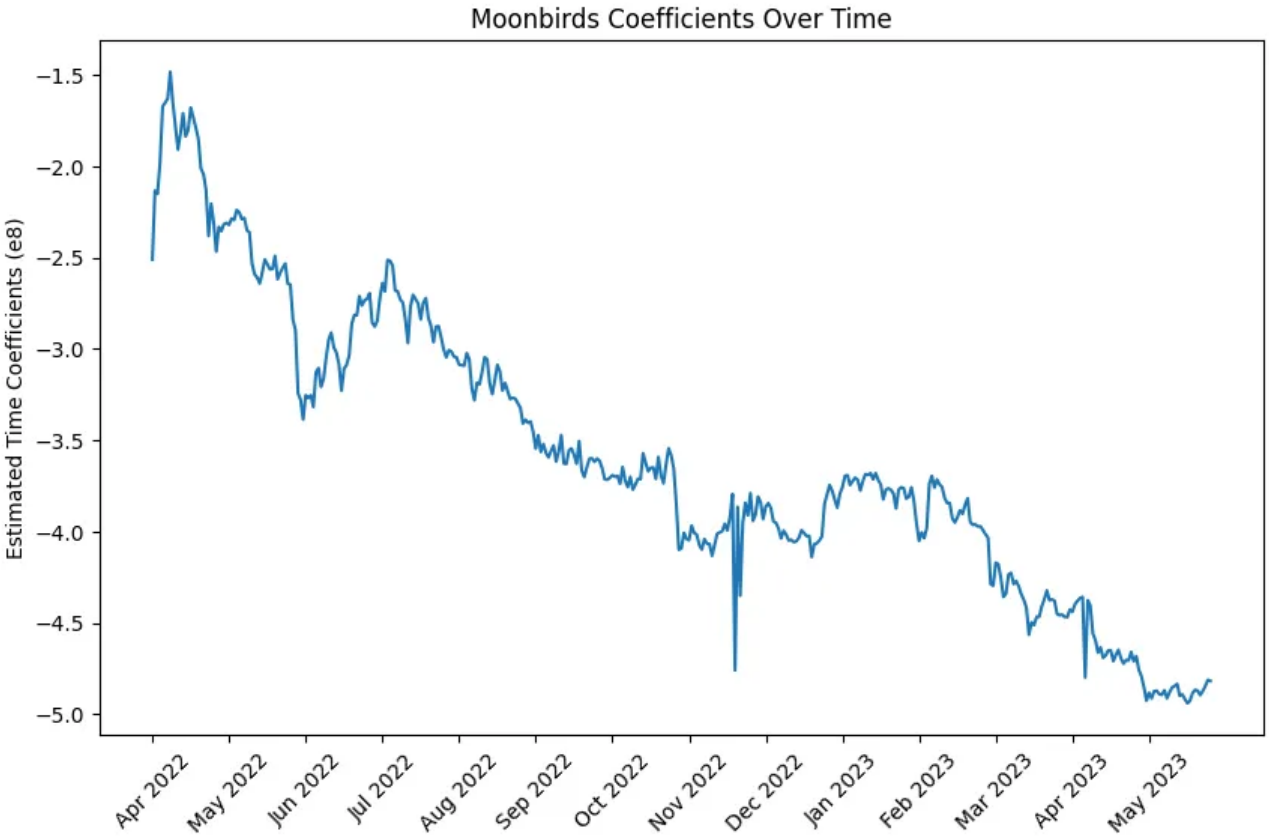


Figure 17: Moonbirds estimated temporal coefficients from April 2022 to May 2023

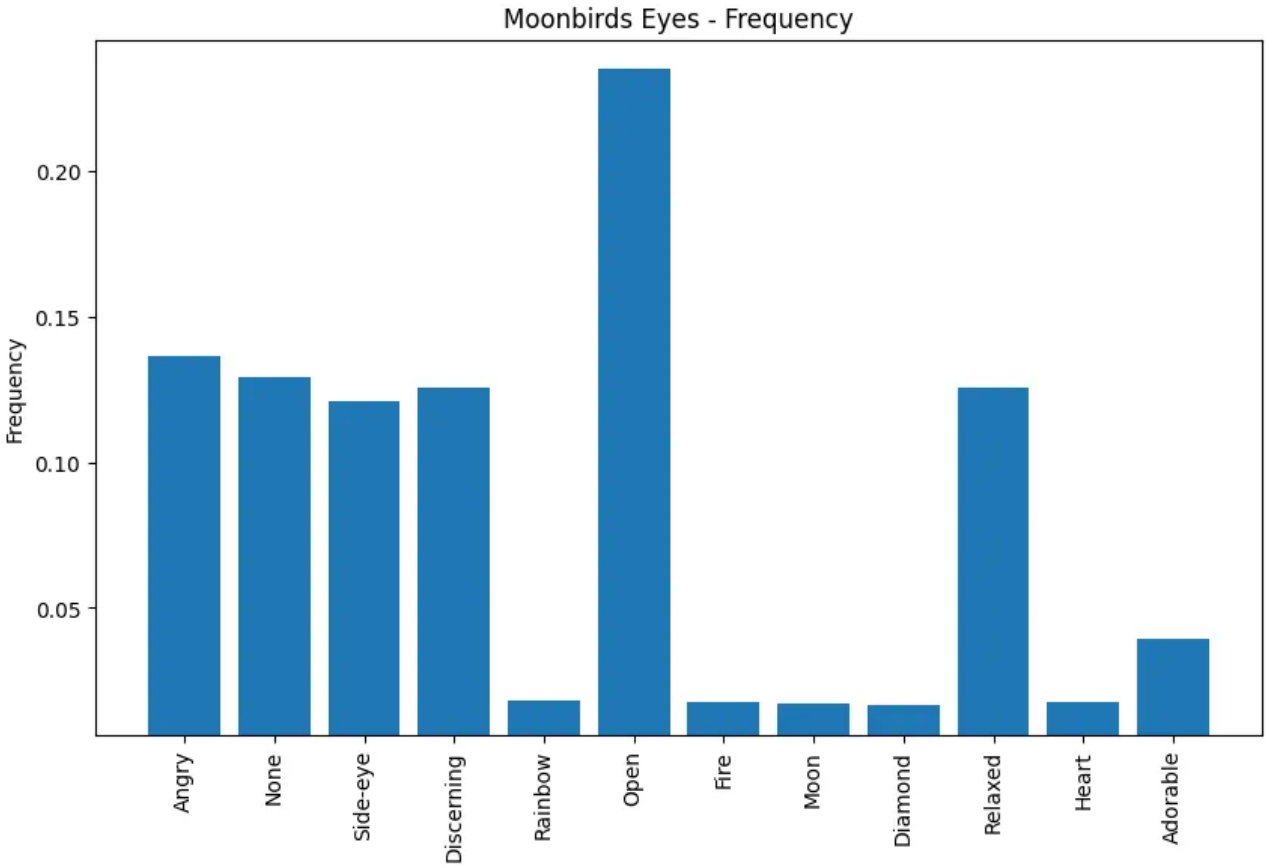


Figure 18: Frequency of eyes traits within the Moonbirds collection

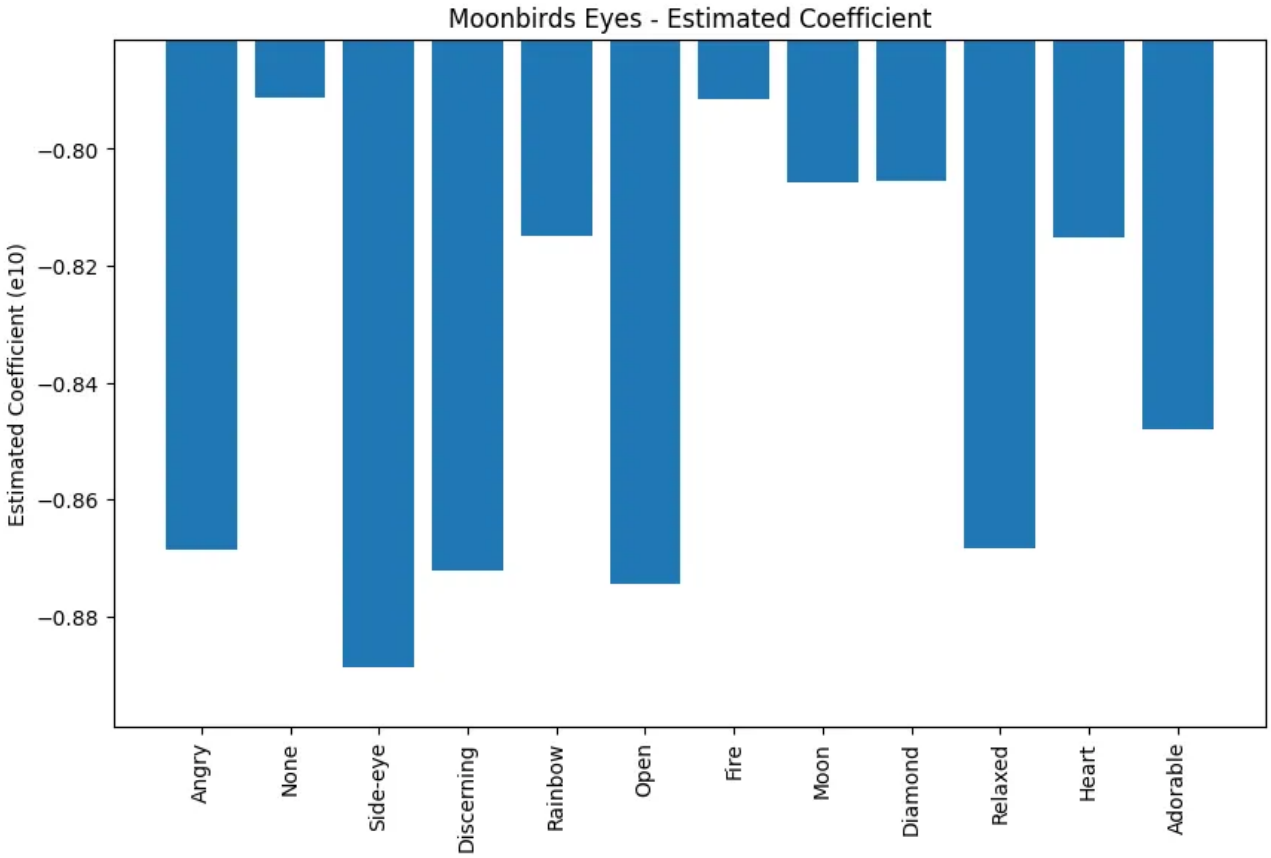


Figure 19: Eyes estimated coefficients within the Moonbirds collection

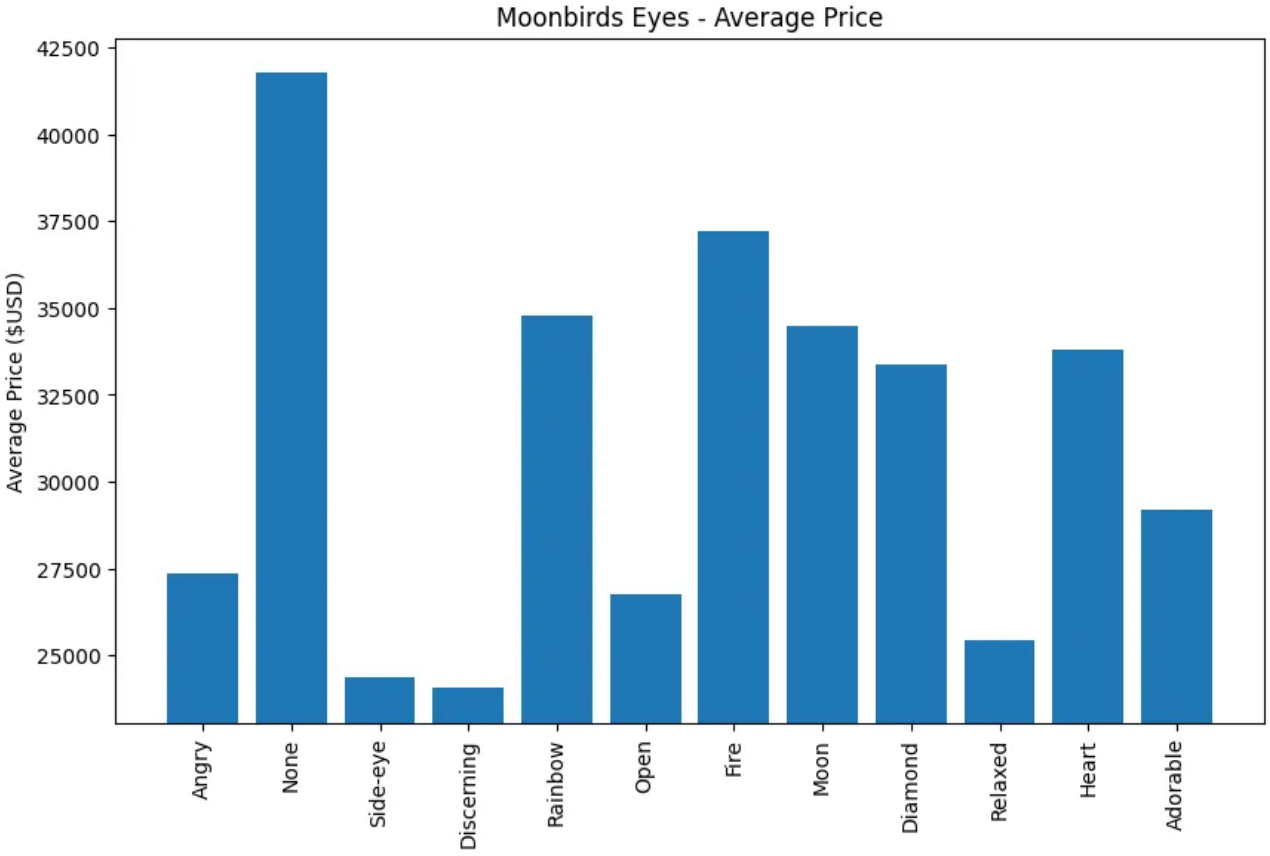


Figure 20: Average sale price of different types of eyes within the Moonbirds collection

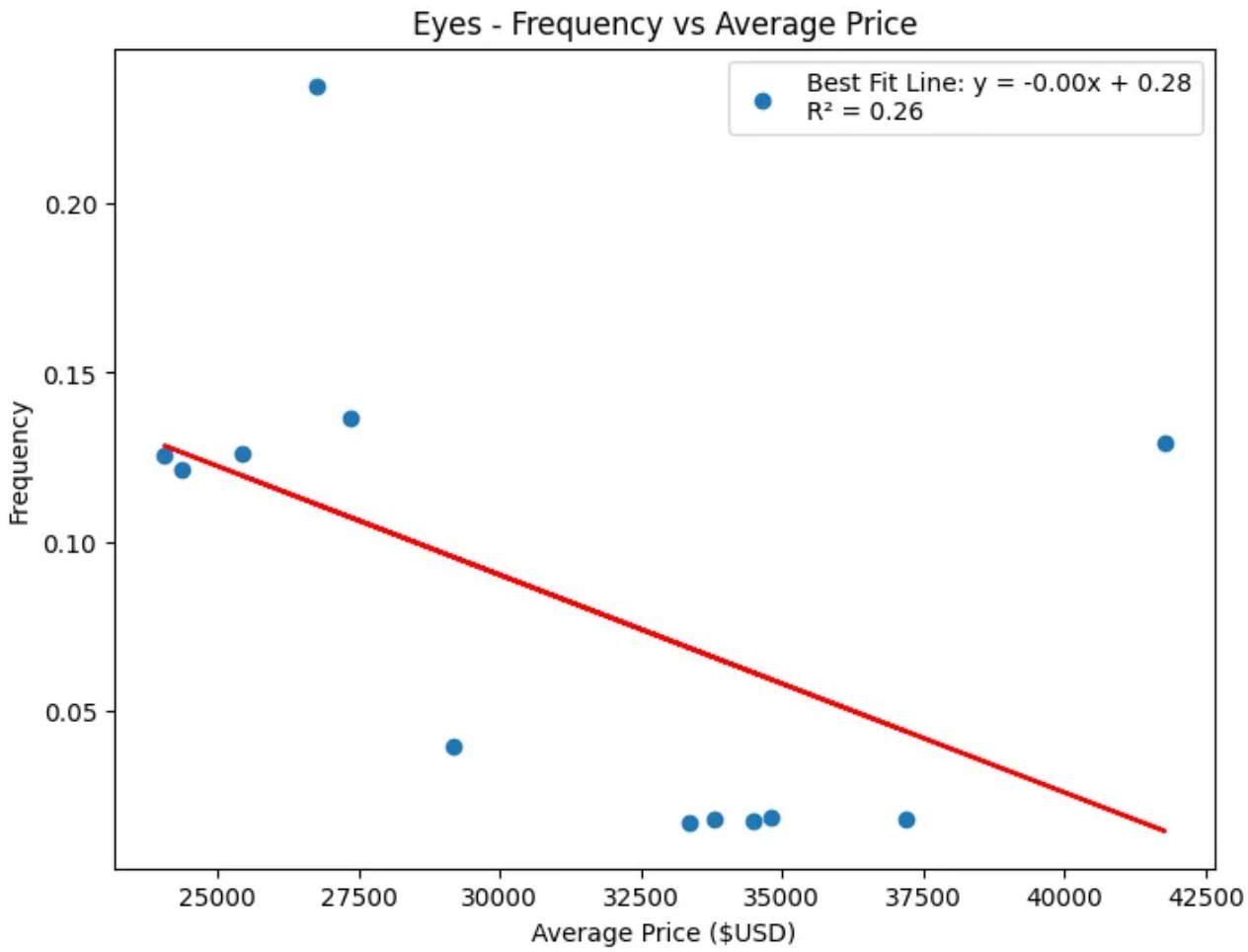


Figure 21: Linear regression of the eyes frequency on the eyes average price

Apart from the individual analyses of each collection, we also conducted a comparative assessment to derive general insights. The results are presented in the following six charts:

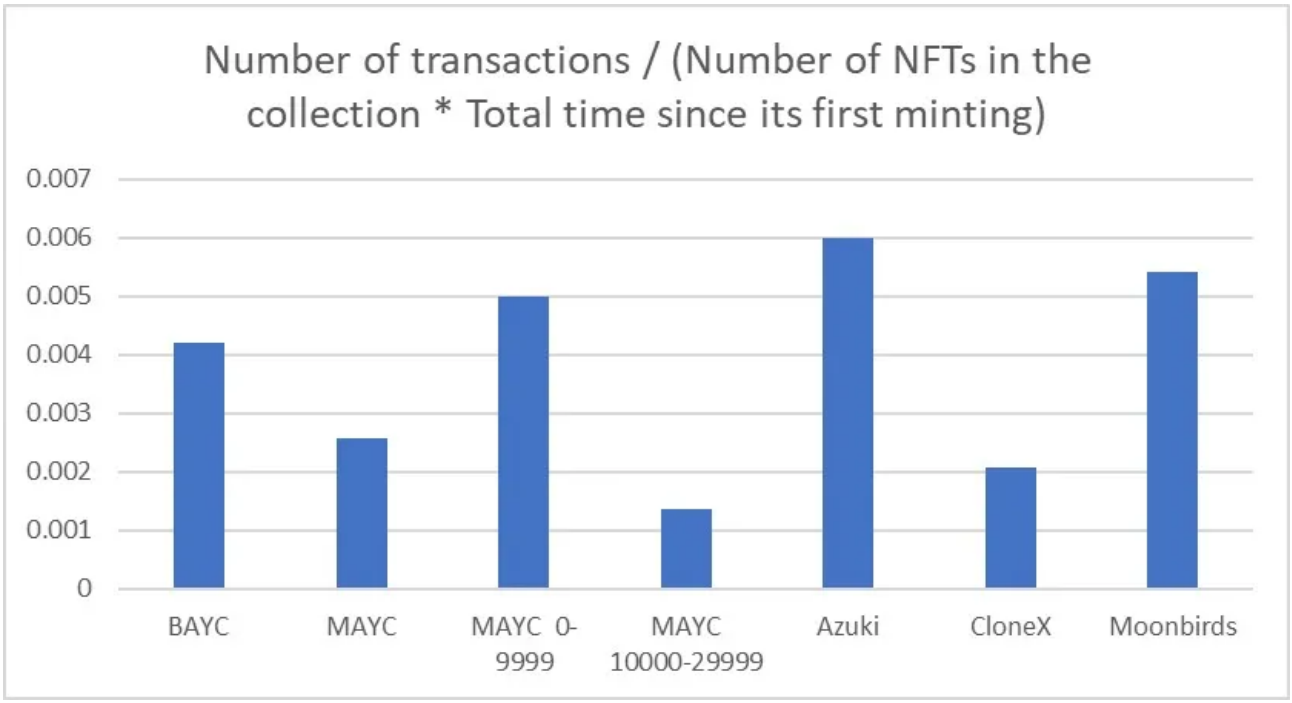


Figure 22: Number of transactions normalized for each Collection

The first Chart represents the Number of transactions / (number of NFTs in the collection * days since its first minting). This is a good KPI that we proposed to analyze the frequency of transactions in a collection. It is important to notice that if we decided to look only for the number of transactions, the number would not be very meaningful: older and bigger collections tend to have more transactions than newer and smaller ones. Then, with this KPI we can eliminate these factors and normalize the number of transactions.

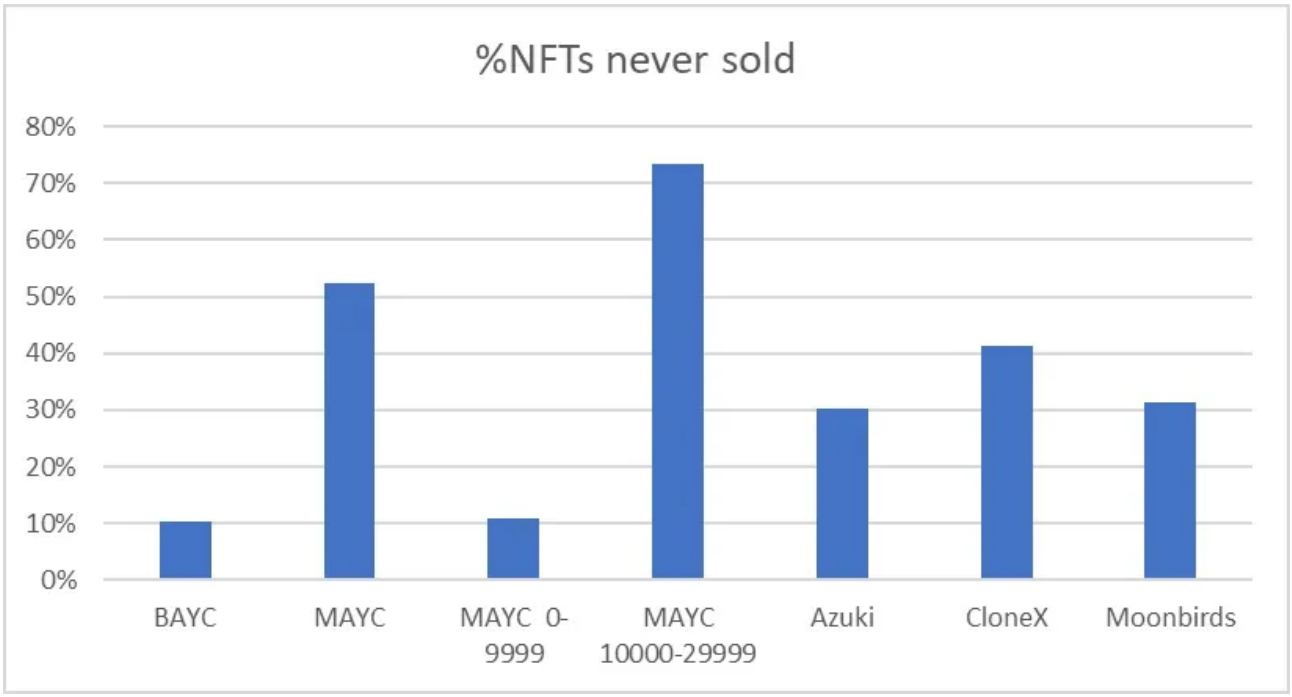


Figure 23: Percentage of NFTs never sold for each collection)

The second chart displays the percentage of NFTs in a collection that were never sold.

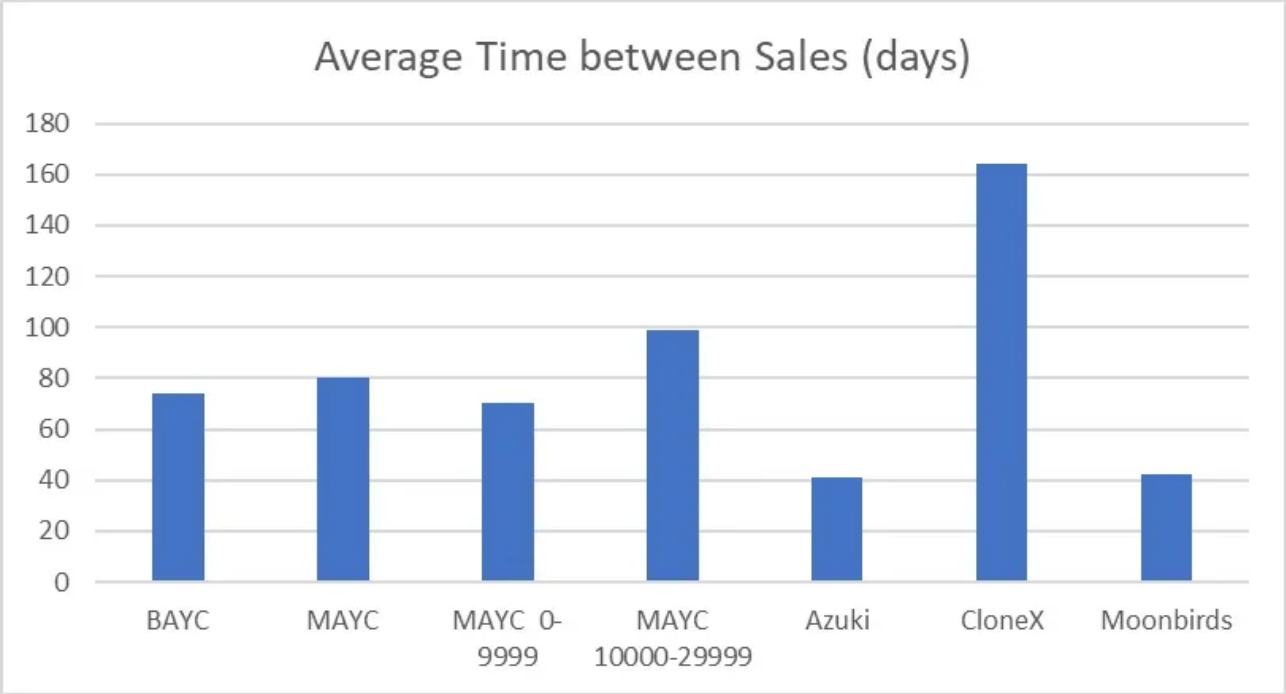


Figure 24: Average time between sales for each collection

In the third chart, we observe the average time in days between sales of the same NFT, including the time of the first sale.

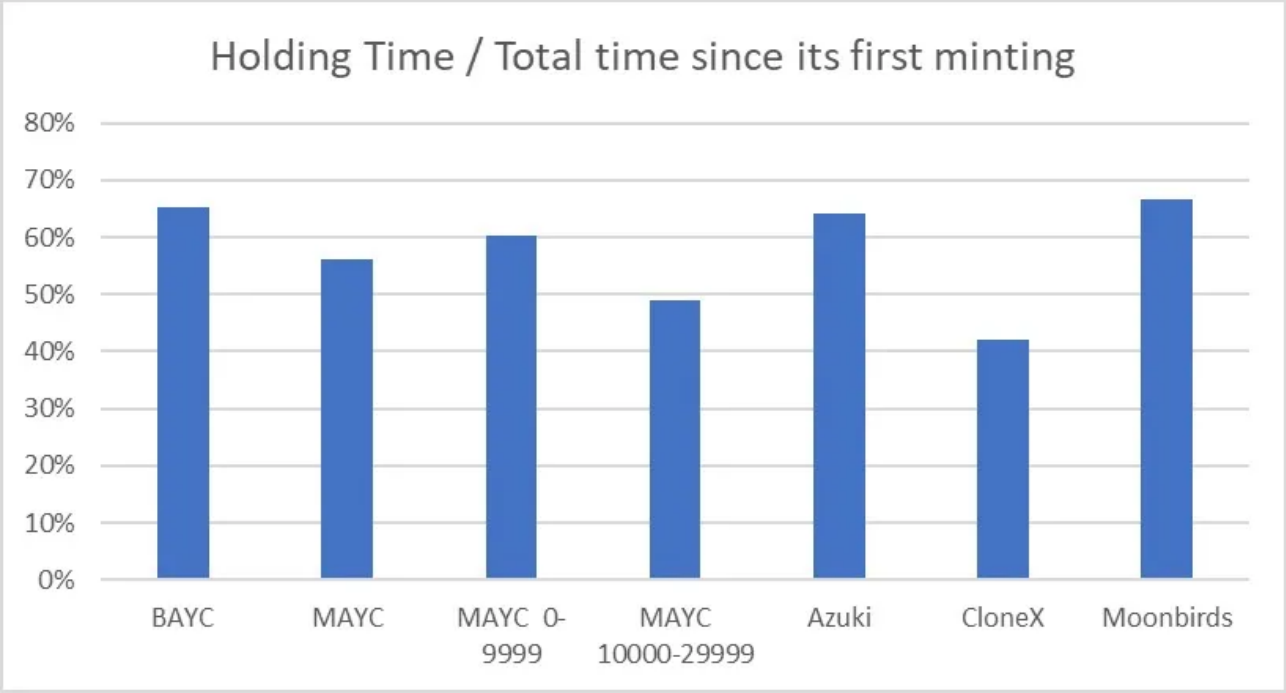


Figure 25: Holding time normalized for each collection

Chart four provides insights into how long people tend to keep their NFT in their wallet. Again in this case, we proposed this KPI which we divide by the total time since its first minting. The reason behind is that older collections tend to have higher holding time than newer ones, because of its age.

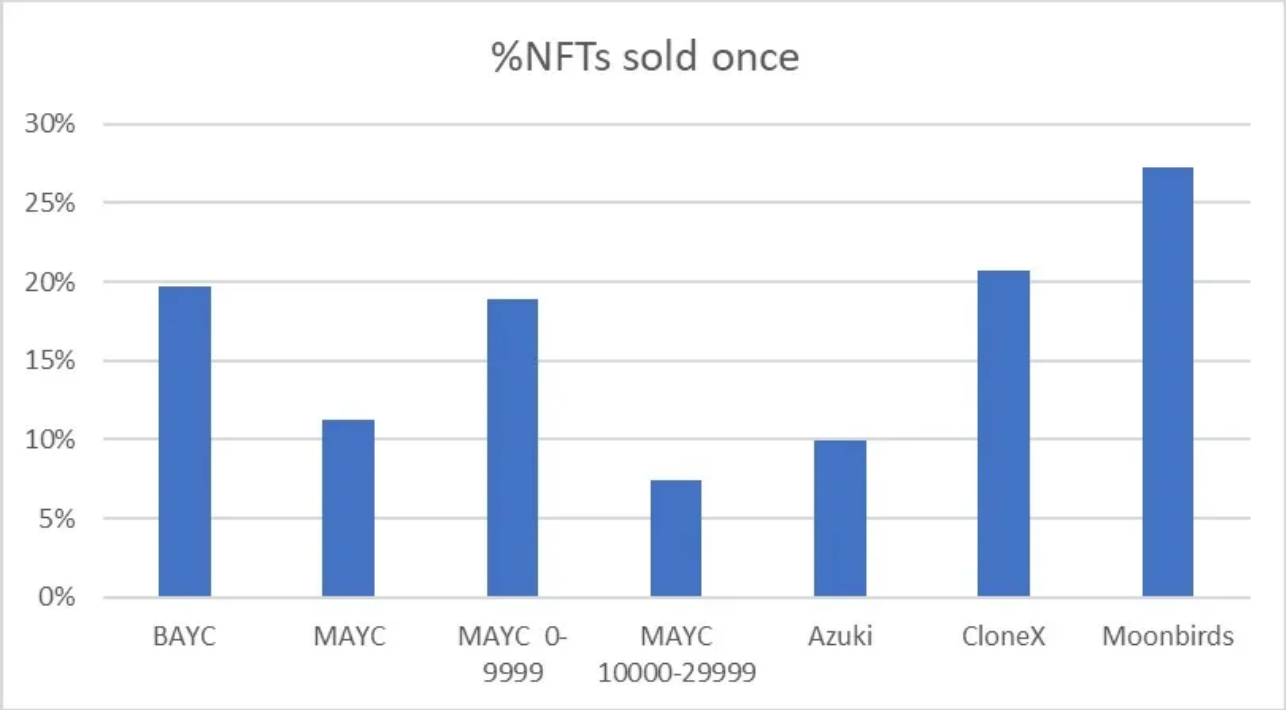


Figure 26: Percentage of NFTs that were sold only once for each collection

The fifth chart shows the percentage of NFTs that were sold only once.

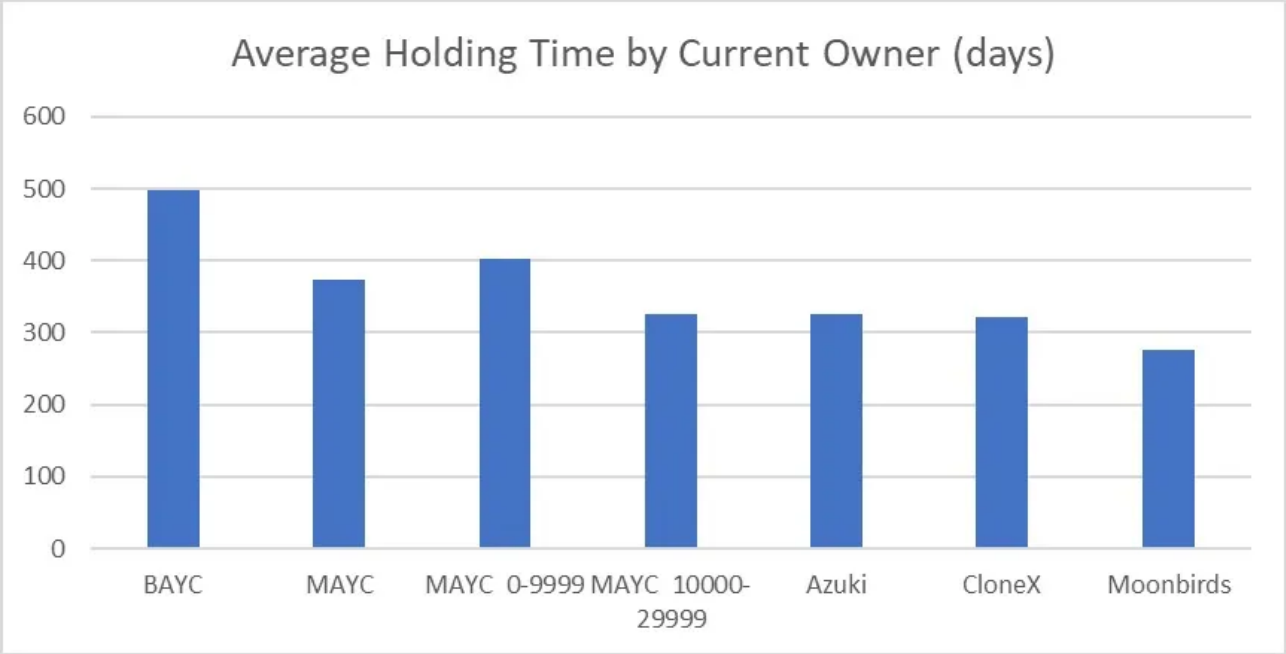


Figure 27: Average holding time from last known sale

Lastly, the sixth chart highlights the holding time, which is the average time that the actual owner of the NFT holds it.

Discussion

The findings from the analysis of the 5 PFP collections have yielded significant insights regarding our research question. They shed light on the characteristics of these collections and the factors that may impact them. Notably, for two collections (BAYC and Moonbirds), traits and aesthetics emerged as influential factors, suggesting that potential buyers consider the visual appeal of the PFPs before making a purchase decision. On the other hand, for three collections (BAYC, Azuki, and Moonbirds), external factors played a substantial role in investors' decision-making process. Lastly, the results for two collections (CloneX and MAYC) were inconclusive, warranting further investigation.

Bored Ape Yacht Club and Moonbirds

Starting with Bored Ape Yacht Club (BAYC) and Moonbirds, the results indicate that the feature vector of estimated coefficients plays a significant role in influencing the price of NFTs, suggesting that potential buyers prioritize the features and aesthetics of an NFT before making a purchase.

Moreover, when considering the percentage of total time that these two collections were held (Figure 25), they stood out among the others. This reinforces the idea that people value these NFTs and tend to hold them for more extended periods because of personal preferences rather than speculative motives. Notably, the data showed that Moonbirds is the collection with the highest percentage of NFTs sold only once (Figure 26), exceeding 25%. On the other hand, BAYC is the collection where people hold onto their NFTs for the longest period of time (Figure 27). These results provide further support to the notion that individuals perceive NFTs from these collections as personal items of value rather than mere speculative assets.

Looking deeper into the traits effects, we selected one trait from each collection to analyze its rarity influence. For the Bored Ape Yacht Club collection, we focused on the "FUR" trait. In figure 5, our findings revealed that the "Solid Gold Fur" is the rarest feature, accounting for less than 2% of the total NFTs in the collection. Additionally, when exploring the average sale price of NFTs in a determined type of fur (Figure 7), the Solid Gold Fur emerged as the most expensive by a significant margin.

We attempted then to establish a linear relationship between rarity and price (Figure 8), but the results were not highly conclusive ($R2 = 0.16$). That result

was led by some cases like the “Blue Fur”, which exhibited an average frequency within the sample, and surprisingly possessed the lowest average price. This complexity in the relationship between rarity and price hindered a strong linear approximation.

Interestingly, all the estimated coefficients from our analysis were positive, indicating that the presence of any trait, regardless of which one, would positively impact the predicted price of the NFT. This suggests that having a particular trait, irrespective of its rarity, contributes to higher price predictions for the NFT.

For the Moonbirds collection, we focused on the “eyes” trait to analyze the relationship between rarity and price. Again in this case, we observed a visual relationship between rarity and price, specifically, the “rainbow”, “fire”, “moon”, “diamond”, and “heart” eyes were the scarcest within the collection (Figure 18) and consequently, we found that NFTs featuring these unique eyes traits, along with the “none” category, commanded the highest prices (Figure 20). However, trying to find a linear relationship to elucidate that relationship was not very explanatory (Figure 21 shows R2 = 0.26).

Intriguingly, the behavior of the estimated coefficients differed from what we found for the BAYC collection. For Moonbirds eyes, eyewear, headwear, and outerwear, the coefficients were negative, indicating a downward pressure on the predicted price. Conversely, the beak coefficients were all positives, suggesting a positive influence on the price prediction. The coefficients for background, body, and feathers were a mix of positive and negative values. For example, legendary guardian, legendary emperor, legendary professor, legendary crescent, legendary brave, and legendary sage feathers had positive coefficients, indicating a preference for these unique features. On the other hand, feathers in colors like gray, blue, white, purple, black, red, brown, prink, green, and none showed negative coefficients, implying a lesser price impact, possibly due to their more common occurrence.

Overall, that interaction between price and traits in the Moonbirds collections was very intriguing, as we exemplified with the eyes coefficients that were all negatives, including the “none”, meaning that with or without this trait, the price would decrease. This finding wasn’t entirely clear in our analysis and warrants further investigation to fully grasp the underlying factors involved in it.

In our model, we not only identified the statistical significance of the feature coefficients, but we also found that the temporal coefficients were significant. After subjecting these coefficients to the Pearson correlation test, we observed a strong correlation relationship between them and the three series we chose to analyze: the ETH price, the S&P 500 index, and the Fed Rate. With the ETH price, it is possible to see that the behavior of BAYC and Moonbids temporal coefficients are similar to the ETH price behavior during certain "crypto crashes" like the Luna crash in May 2022 and the FTX crash in November 2022.

However, it is essential to emphasize that correlation does not imply causation, and we cannot establish a direct causal link based on these correlations. Furthermore, we found correlations with two additional macroeconomic variables: the S&P5500 and the Fed Rate. This indicates that movements beyond the NFT environment exhibit similar behavior to that observed within these collections. While the exact causal mechanisms remain uncertain, these correlations highlight the interconnectedness between broader economic conditions and NFT pricing.

During the Granger causality test, for the BAYC collection, the analysis yielded no evidence of causality between the temporal coefficients and the time series proposed, even when exploring different time lags. In contrast, for the Moonbirds collection, the causality test indicated a significant causation between the temporal coefficient and the ETH price, with a lag of 5 time periods (Table 9). This implies that fluctuations in the ETH price play a contributory role in shaping price variations within the Moonbirds collection.

This result is in line with (Ante 2021), who used data between January 2018 and April 2021 to demonstrate the impact of cryptocurrency markets on the NFT market, finding that NFT sales were influenced by Bitcoin and Ether price shocks. Furthermore, (Dowling 2021b) reinforces this result, as his exploration of the NFT market in early 2021 suggested that cryptocurrency pricing may provide valuable insights into NFT pricing patterns.

However, it is important to approach these findings with a degree of caution. Causality in complex systems like the NFT market can be influenced by numerous interrelated factors, and the precise mechanisms at play may require further investigation.

Azuki

In our investigation of the Azuki collection, we made intriguing discoveries regarding the significance of feature and temporal coefficients. Different from the first two collections, the feature coefficients were not statistically significant, indicating that within this collection, people seem to be less concerned about specific NFT features and more focused on market conditions and various internal and external factors. This observation suggests that the Azuki collection may lean towards being a speculative collection, where buyers are driven by investment opportunities rather than individual NFT characteristics.

Supporting this notion, Figure 24 reveals that Azuki exhibits the lowest average time between sales, 41 days, suggesting that individuals tend to buy and sell Azuki NFTs quickly, treating them more like short-term investments rather than holding them for extended periods. In addition to that, we proposed a new KPI that also corroborates with that result (Figure 22): number of transactions per number of nfts in the collection per day since its first minting. Azuki was on top of the other collections in that chart, showing its high trade frequency.

As we did not observe any statistically significant feature coefficients, we did not explore their relationship with the rarity of traits within the Azuki collection. Instead, we proceeded with correlation and causality tests to delve further into the dynamics at play.

During our analysis, we discovered correlations between the three series we examined with the temporal coefficients in the Azuki collection. Additionally, the Granger causality test revealed that the S&P500 at lag 2,3,4,or 5, exhibited a significant correlation with the Azuki NFT price. These findings align with our previous observations, suggesting that investors may closely monitor the behavior of the S&P500 to inform their investment decisions concerning specific Azuki NFTs. Consequently, movements in the S&P500 are likely to influence the overall price of the Azuki collection.

This connection between a market indicator and the Azuki NFT price highlights the speculative nature of the collection, where external market factors hold sway over the NFT pricing dynamics.
That result diverged from the findings by (Aharon and Demir 2021), who investigated NFTs' connectedness with other financial assets during COVID-19, concluding that NFTs have an independence from common asset classes (equities, bond, currencies, gold, and oil).

The observed divergence in findings may be attributed to the difference in the time frames analyzed. Aharon and Demir studied the period from January 2018 to June 2021, while Azuki's first mint occurred in 2022. As the NFT market rapidly evolves, early stages might not reflect its current state and hence, the NFT landscape at the beginning of the study might differ significantly from the present scenario. Therefore, considering the time gap is crucial to account for the dynamic nature of the NFT market.

CloneX

Our next examination was CloneX and it posed unique challenges different from the other collections. Regrettably, we did not find any statistically significant coefficients in our model. This suggests that the model we proposed might not be the most suitable for this particular collection.

One notable aspect of CloneX is its status as the oldest collection among all those we studied, with over two years since its first minting. However, surprisingly, approximately 40% of CloneX NFTs remain unsold (Figure 23). This significant number of unsold NFTs introduces complexities that impact the model's precision in capturing all influences, particularly concerning the features coefficients. Since only 60% of the NFTs were sold at least once, the model only considers the features of those sold NFTs, leaving some traits that were never sold unaccounted for. This limitation has a detrimental effect on the reliability of the coefficients within the model.

Regarding temporal coefficients, we observed that CloneX exhibited the highest value in the Average Time Between Sales, reaching 164 days (Figure 24) . This implies that CloneX NFTs are not traded frequently, making it challenging to incorporate the temporal aspect into the model due to the scarcity of sales data.

Given these limitations and the lack of sufficient statistically significant coefficients, we made the decision to not pursue further tests of correlation and causality. Instead, we acknowledge the uniqueness of the CloneX collection and the difficulties it presents in analyzing its pricing dynamics.

Mutant Ape Yacht Club

MAYC stands out as another distinct collection that posed its own unique

challenges during our analysis. Running the model for all NFTs within the collection (token ID from 0 to 29999), did not yield any statistically significant coefficients. Similar to CloneX, MAYC's peculiarities influenced these results.

One notable aspect of MAYC is that over 50% of its NFTs were never sold (Figure 23), surpassing even the percentage observed in CloneX. Consequently,certain traits within MAYC were never considered in our model, impacting the precision of feature coefficients. This lack of sales data for a significant portion of the collection introduced complexities in the comprehension of the price dynamics.

Furthermore, the average time between sales for Mutant Ape Yacht Club NFTs is 80 days (Figure 24), outstanding the duration observed in BAYC, Azuki, and Moonbirds collections. That infrequency of sales further complicated the inclusion of temporal effects in the model, making it challenging to capture the dynamics of pricing fluctuations.

Given these challenges, our initial analysis, a naive one, faced limitations in fully grasping the pricing behavior within the MAYC collection. However, acknowledging these unique characteristics motivates us to explore alternative approaches and refine our methodology.

We made a deliberate division, creating two distinct sub-collections.The first collection comprises token IDs ranging from 0 to 9999, while the second collection includes token IDs from10 to 29999. The division is not arbitrary; it aligns with significant differences in the minting process and requirements for each group.

The first 10000 MAYC NFTs were generated through the usual process of AI image generator, where a random vector of features creates unique NFTs that can be minted by anyone in the community. This approach is consistent with the typical NFT minting process, making this subset a representative sample of conventional NFTs within the MAYC collection.

The second group, however, follows a distinctive protocol for minting. To acquire a Mutant Ape Yacht Club NFT from this subset, individuals need to possess a Bored Ape Yacht Club NFT, and additionally, they require a special element called "Serum", available in two types: number 1 and number 2. With these prerequisites met, individuals can mint their MAYC NFT, whose process will follow the same steps as the first group, through an AI image generator.

This strategic division enabled us to examine two distinct categories within the MAYC collection, shedding light on the contrasting dynamics between conventional minting and the exclusive requirements for minting within the second group.

During our analysis of the first part of the MAYC collection (token IDs 0-9999), we encountered an unexpected result. Despite a high percentage of NFTs sold, Figure 23 shows only 11% of NFTs in that group remained unsold, a considerable number of transactions per NFT per time of existence (Figure 22), and a low average time between sales (Figure 24), we did not find any statistically significant coefficient.

These characteristics led us to expect a similar behavior to that of BAYC and Moonbirds collections, where the feature and temporal coefficients significantly influenced price predictions. However, as we have seen, this was not the case.

At this point, we have not yet determined the reasons behind this outcome, and further investigation is needed to understand the implications fully.
On the other hand, the second part of MAYC collection (token IDs 10000-29999) exhibited a behavior more akin to the BAYC and Moonbirds collections.

These NFTs stand out as unique with its way of minting, fostering a sense of pertinence among collectors that mint it. As a result, a significant portion of these MAYC NFTs were never sold (over 70% according to Figure 23).

In addition to that, Figure 22 reveals infrequent trading activity for these NFTs, suggesting that collectors tend to hold onto their NFTs and this sub-collection has the second-highest average time between sales (Figure 24), behind CloneX. This aspect emphasizes the collector-oriented nature of the owners of these NFTs. Despite all these characteristics, we still observed some temporal influence in the predicted price (there were some significant temporal coefficients in our model), indicating that those who decide to sell their minted NFTs may consider market conditions in their decision-making.

Due to the absence of a continuous time series for our temporal coefficients, with some coefficients turning out to be statistically insignificant, we made the decision to not proceed with the correlation and causality tests. Similarly, we encountered some features coefficients that lacked significance, resulting in missing coefficients. As a consequence, we opted against verifying the relationship between the rarity of traits and their estimated coefficients.

The limitations posed by these missing coefficients in both temporal and feature coefficients prevented us from conducting further analyses to explore correlations and causal relationships. Recognizing these constraints, we acknowledge the need for a more comprehensive dataset to gain deeper insights into the dynamics and influences governing the NFT pricing within the collection.

In conclusion, the results emphasized that the aesthetics and features of NFTs have a significant impact on the pricing of BAYC and moonbirds, and people tend to hold NFTs they appreciate for more extended periods. The Moonbirds collection, despite having fewer transactions overall, witnessed a substantial percentage of NFTs sold only once, showcasing the initial intention of buyers to retain these items in their wallets. The Azuki collection appeared as a speculative collection, with external market factors influencing its pricing dynamics more than specific NFT features. CloneX and MAYC presented unique challenges in analyzing their pricing behavior due to their particularities and their data availability.

Conclusion

In this research, we conducted a comprehensive analysis of five PFP collections, identifying the factors influencing the pricing dynamics of these digital assets. Through rigorous examination, we made significant findings that contributed to our understanding of the NFT market.

Firstly, we discovered that certain collections possess a speculative factor, wherein the NFT prices are influenced by external market conditions. For instance, the Azuki collection demonstrated a strong correlation between its NFT prices and the S&P 500 index, indicating a close relationship between the broader economy and the pricing dynamics of these NFTs. On the other hand, Moonbirds exhibited a causal relationship with fluctuations in the ETH price, suggesting that market movements in the Ethereum cryptocurrency play a contributory role in shaping price variations within the collection.

Secondly, we observed that some collections had an appeal factor, where NFT prices were influenced by intrinsic factors such as aesthetics features. Notably, Bored Ape Yacht Club and Moonbirds collections showed correlation between specific NFT features and their prices, indicating that potential buyers prioritize visual appeal and unique traits before making a purchase decision.

In all cases where a temporal effect was significant in the model, the ETH price, S&P500, and Fed Rate exhibited correlations with the temporal coefficients estimated.

However, while our research yielded valuable insights for some collections, we also encountered challenges in constructing a precise model for the Mutant Ape Yacht Club and CloneX collections. The complexity and uniqueness of these collections hindered a robust analysis. As a result, further investigation is required to provide explanations for the difficulties encountered and to gain deeper insights into the dynamics driving the pricing behavior within these collections.

Moving forward, future research could apply the repeat sales method to the Azuki collection, given its speculative nature, and also for MAYC and CloneX, trying to identify the underlying pricing patterns and relationships.
In addition to traditional econometric approaches, future research papers could employ machine learning models to analyze NFT pricing dynamics. Leveraging machine learning algorithms could help identify complex patterns and relationships between NFT characteristics, external factors, and transaction prices.

Another approach for exploration is the influence of social media platforms on NFT pricing and price volatility. Treating NFT price volatility as a social phenomenon rather than merely a mathematical or economic problem would provide a more comprehensive understanding of NFT market behavior and the role of social influence.

Lastly, expanding the research to explore other NFT categories beyond PFPs is essential for gaining a holistic view of the NFT market. Each category may possess distinct characteristics, driving factors, and market dynamics. Conducting similar analyses for other categories, such as virtual real estate, digital art, and gaming NFTs, would enrich the overall understanding of the entire NFT ecosystem. By broadening the scope of the study, we can capture the diversity and complexities of NFT markets, contributing to a more comprehensive knowledge base for the digital asset space.

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